

BOUNDARY-RAY FACTORIZATION AND GEOMETRIC WALL-CROSSING IN OPEN FJRW THEORY OF $x^r + y^s$

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ABSTRACT. Gross, Kelly and Tessler constructed genus-zero open FJRW invariants for $(x^r + y^s, \mu_r \times \mu_s)$ and a group of automorphisms describing their dependence on boundary conditions. We prove that every primary transition factors uniquely as an increasing-slope product over the rays spanned by the two boundary-marking counts of its critical graphs. With descendents, the same holds after restriction to any finite divisor-closed set of deformation monomials. Two distinguished systems, characterized by the survival of one extreme coefficient in each family of balanced-disk coefficients joined by critical graphs, determine a canonical boundary-to-boundary factorization. Every open potential is recovered from unique initial coefficients by its successive ray factors.

For every finite primary or descendent coefficient set, an actual concatenation of Gross–Kelly–Tessler canonical homotopies realizes this factorization: after projection, each stage has nonzero jumps on one ray, and the stages occur in increasing slope. The first quadrant of boundary-marking counts, with its origin removed, is a radial half-line times the interval of slopes. Hence the finite crossing product detects only slope: radial paths act trivially, and the total product is boundary-to-boundary transport rather than origin monodromy. These continuation data alone therefore do not produce a singular integral-affine base. Exact calculations for $x^5 + y^5$ give the first ray factors and new invariants, verify the open topological recursion, and isolate both a factor-of-two discrepancy in a published example and the omission of the torus direction $x\partial_x - y\partial_y$ from the stated wall-crossing algebra.

1. INTRODUCTION

1.1. **The theory.** Gross, Kelly and Tessler [GKT22] defined genus-zero open FJRW invariants

$$\nu_{\Gamma, \mathbf{d}} = \left\langle \prod_i \tau_{d_i}^{(a_i, b_i)} \sigma_1^{k_1} \sigma_2^{k_2} \sigma_{12} \right\rangle^{s, o}$$

for the Landau–Ginzburg model $(W = x^r + y^s, \mu_r \times \mu_s)$: relative Euler numbers of descendent Witten bundles over moduli of W -spin disks, which are real orbifolds with corners. The invariants depend on a family $\mathfrak{s}$ of canonical multisections (boundary conditions). Three of their results organize this dependence completely. First, the systems of invariants arising from canonical families form a set $\Omega_{r,s}(I)$ (I a finite set of internal marking labels as in [GKT22]; in universal finite-order statements below it is chosen *saturated* at the order under consideration, meaning it carries at least $k - 1$ labels of every primary twist at order k), characterized by explicit polynomial conditions (the *chamber-index* conditions of [GKT22, Def. 3.47, Cor. 5.14]; a *chamber index* is their name for such a system of invariants, not a connected component of the ray complement in this paper): normalizations for at most one internal marking, and the conditions $\mathcal{A}(J, \mathbf{d}, \nu) = (-1)^d$ (singletons) and $\mathcal{A}(J, \mathbf{d}, \nu) = 0$ for $d(J, \mathbf{d}) < 0$ (with $d(J, \mathbf{d})$ as in

Date: August 24, 2026.

2020 Mathematics Subject Classification. 14N35 (primary); 14J33, 14T90, 53D45 (secondary).

Key words and phrases. Open FJRW theory, wall-crossing, tropical vertex, Landau–Ginzburg models, boundary conditions.

[GKT22, Def. 4.8]), where $\mathcal{A}$ is a Γ -function-weighted polynomial in the invariants. Second, Gross–Kelly–Tessler prove that their wall-crossing group acts faithfully and transitively on $\Omega_{r,s}(I)$ [GKT22, Thm. 4.27]; Definition 4.23 of [GKT22] includes one generating summand for each critical graph of nonzero bidegree. The marginal bidegree- $(0,0)$ case, which is omitted from that definition but appears in the jump and realization arguments, is treated separately below. Third, the generating-function potential W^ν of any system has oscillatory integrals over the good basis equal to the closed extended FJRW generating function [GKT22, Thm. 0.3], an open mirror theorem building on the closed Landau–Ginzburg mirror symmetry of [LLSS17, HLSW22].

Gross–Kelly–Tessler remark [GKT22, Rem. 0.8] that their wall-crossing group is close in spirit to the tropical vertex group of Kontsevich–Soibelman [KS06], as used for Fano mirror symmetry in [GPS10, Gro10] and in the log Calabi–Yau setting [GHK15], and their survey raises the problem of a tropical formulation of the open theory [GKT26]. Such a formulation is not immediate. A Landau–Ginzburg model carries no target space and hence no SYZ fibration, and on these grounds Maher [Mah24, Rem. 5.1.1] records the combinatorial machinery of the Gross–Siebert program as out of reach for these models. The present paper develops a slope-ordered boundary-ray factorization of the wall-crossing group and computes its first factors. The construction is algebraic and does not presuppose a degenerating target. Every finite primary or descendent coefficient window is realized by canonical homotopies in Theorem 1.6. The algebraic ray factors, but not the chosen homotopies, are compatible in the coefficientwise limit. The sector itself is the effective cone of the two open-FJRW boundary charges. Theorem 1.7 proves that the induced continuation theory on its punctured cone factors through slope and has no origin loop. Thus the cone organizes boundary-to-boundary wall-crossing, but a singular affine position space of the kind supplied by an SYZ fibration requires genuinely additional input.

1.2. Results. Throughout, $\mathfrak{g}^{r,s}$ denotes the Landau–Ginzburg wall-crossing Lie algebra of [GKT22, Def. 4.23]: it is spanned by elements $u_{J,\mathbf{d}}X_{k_1,k_2}$ with

$$X_{k_1,k_2} = x^{k_1}y^{k_2}((k_2+1)x\partial_x - (k_1+1)y\partial_y)$$

the Hamiltonian vector field of $x^{k_1+1}y^{k_2+1}$ for $dx \wedge dy$, and $u_{J,\mathbf{d}} = \prod_j t_{a_j,b_j,d_j}$ a monomial in the deformation parameters ($t_{a,b,d}$ dual to the insertion $\tau_d^{(a,b)}$; the descendent index is suppressed when $d = 0$), subject to the congruences $k_1 \equiv r(J) \pmod{r}$, $k_2 \equiv s(J) \pmod{s}$ and the dimension constraint

$$(1) \quad sk_1 + rk_2 = m(J, \mathbf{d}) - rs, \quad m(J, \mathbf{d}) = rs + \sum_j (sa_j + rb_j + rs(d_j - 1)).$$

The right-hand side of (1) is the *Euler weight* $w(J, \mathbf{d})$ of the monomial $u_{J,\mathbf{d}}$. Write

$$\sum_{j \in J} a_j = r(J) + r\ell_1, \quad 0 \leq r(J) < r, \quad \sum_{j \in J} b_j = s(J) + s\ell_2, \quad 0 \leq s(J) < s,$$

and set

$$N(J, \mathbf{d}) = \ell_1 + \ell_2 - |J| + 1 + \sum_{j \in J} d_j.$$

By [GKT22, Not. 2.40], when $N \geq 0$ there are $N+1$ balanced disk types (potential coefficients), and when $N \geq 1$ there are N critical graphs; below the respective thresholds those sets are empty. The Lie bidegree of a critical graph is its boundary degree shifted by $(-1, -1)$: the wall Hamiltonian $x^{k_1+1}y^{k_2+1}$ is the boundary monomial $\sigma_1^\# \sigma_2^\#$ of the critical disk, and the shift is the removal of the single fully twisted point σ_{12} that balanced disks carry and critical disks do not. This is the primitive form $dx \wedge dy$ acting as an integral-affine shift (§2).

We include the bidegree- $(0, 0)$ generators that occur when $w = 0$. Thus

$$\tilde{\mathfrak{g}}^{r,s} = \mathfrak{g}^{r,s} \oplus \bigoplus_{\substack{\emptyset \neq J, w(J, \mathbf{d})=0, N(J, \mathbf{d}) \geq 1 \\ r(J)=s(J)=0}} \mathbb{Q} u_{J, \mathbf{d}} (x \partial_x - y \partial_y), \quad \tilde{G}^{r,s} = \exp(\tilde{\mathfrak{g}}^{r,s}).$$

Remark 2.3 proves that this is a Lie algebra and explains its relation to GKT's stated algebra.

Here are the conventions needed to read the theorem statements. A *chamber index* is GKT's name for a system of open invariants satisfying the polynomial conditions of [GKT22, Def. 3.47]; by [GKT22, Cor. 5.14], these are exactly the systems realized by canonical multisection families. A *simple canonical homotopy* is a one-parameter family of such boundary conditions in the normal form of [GKT22, Lem. 3.40]; its isolated critical zeros produce the jump automorphisms in the proof of [GKT22, Thm. 5.9]. A finite set S of deformation monomials is *divisor-closed* if every monomial dividing an element of S also belongs to S ; reduction to S means quotienting out all other deformation coefficients.

For fixed internal data $(J, \mathbf{d})$, the corresponding $N + 1$ balanced potential coefficients form the $(J, \mathbf{d})$ -*diagonal*; they are consecutive lattice points on a line, and the N intervening critical graphs act on adjacent coefficients. The *boundary ray* of a generator $u_{J, \mathbf{d}} X_{k_1, k_2}$ is $\mathbb{R}_{\geq 0}(k_1 + 1, k_2 + 1)$. The *direction sector* is the closed first quadrant spanned by the two boundary-marking counts, and a *slope cut* is a ray in this quadrant, with a choice of side when it coincides with a boundary ray. A *good-basis period functional* is the linear functional on a diagonal obtained by taking the corresponding coefficient of the oscillatory periods in [GKT22, Thm. 0.3]; it is written explicitly in Lemma 4.1.

For a potential monomial, its *anchor cut* is the one-sided cut through its own exponent, chosen toward slope one. The *seed coefficient* is the cut-independent initial coefficient obtained after subtracting all transports from proper divisor monomials; a contribution from another seed is called *foreign*. A *tame path* in the punctured direction sector has off-wall endpoints and meets the finitely many rays nonzero modulo S transversely, except for finitely many generic tangencies. These definitions are repeated with full formulas in Section 7.

Theorem 1.1 (Weight trichotomy and the threshold; §2). *In the enlarged algebra $\tilde{\mathfrak{g}}^{r,s}$ defined above, for a nonempty marking multiset J , wall summands over the monomial $u_{J, \mathbf{d}}$ exist if and only if $N(J, \mathbf{d}) \geq 1$ and $w(J, \mathbf{d}) \geq 0$. The pro-nilpotent summands are exactly those of positive weight. For a pro-nilpotent wall, positive weight is necessary but not sufficient (at $r = s = 5$ the monomial $t_{3,3}$ has $w = 5$ but $N = 0$, hence no wall); at $w = 0$ the constraint (1) admits only the marginal torus bidegree $(0, 0)$, and at $w < 0$ it admits nothing. Consequently a monomial of nonpositive weight carries no pro-nilpotent wall generator of its own (although higher-order transport from divisor monomials may reach its coefficients), and the central charge $\hat{c}_W = 2 - \frac{2}{r} - \frac{2}{s}$ governs a trichotomy: pro-nilpotent $d = 0$ wall-crossing exists if and only if $\hat{c}_W > 1$ (we call the pair (r, s) hyperbolic); at $\hat{c}_W = 1$ (parabolic) primary wall-crossing exists but is carried entirely by the marginal torus direction; for $\hat{c}_W < 1$ (simple) there is no primary wall-crossing at all.*

Theorem 1.2 (Slope-ordered boundary-ray factorization; §3). *(i) (Loop consistency.) For any family of simple canonical homotopies supplied by [GKT22, Lem. 3.40] with equal endpoints, the time-ordered product of the critical-graph jump automorphisms constructed in the proof of [GKT22, Thm. 5.9] is the identity; for such homotopies with arbitrary endpoints, the total product depends only on the endpoints. (ii) (Boundary-ray presentation.) In the primary algebra, modulo every power of the deformation ideal, every element of the enlarged wall-crossing group factors uniquely as a slope-ordered product of exponentials supported on the shifted boundary rays $\mathbb{R}_{\geq 0}(k_1 + 1, k_2 + 1)$; the marginal torus terms belong to the $(1, 1)$ -ray. The*

compatible factorizations define its completed boundary-ray presentation. In the full descendent algebra the same presentation is taken on finite divisor-closed coefficient sets and passed to the coefficientwise limit. We call this the completed slope-ordered boundary-ray factorization. It is boundary-to-boundary transport, not a scattering diagram on a complete affine base, and its total transport need not be the identity.

The statements below use the following language. The *diagonal* of a deformation monomial is its set of balanced coefficients, the lattice points of one line in the boundary-degree plane (Definition 2.1). The rest is developed in §7: the *direction sector* is the closed first quadrant of boundary-monomial directions; a *slope cut* is a direction in the sector together with a choice of side when a wall sits on that direction; the transport $\Psi(q)$ to a cut q is the slope-ordered product of the wall factors from the σ_1 -axis side (the *base cut*) up to q ; and a bending contribution to a coefficient is *foreign* if it originates from another monomial's seed (Definitions 7.1 and 7.4).

Theorem 1.3 (Extremal normal forms and the canonical boundary-ray factorization; §5). *On every wall-bearing diagonal, including a marginal diagonal, the wall action on the $N+1$ potential coefficients is triangular; consequently there exist unique chamber indices $\nu^{\min}$ and $\nu^{\max}$ in which, on every such diagonal, only the extreme coefficient survives. The boundary-slope-ordered factorization of the unique group element carrying $W^{\nu^{\min}}$ to $W^{\nu^{\max}}$ is a canonical, choice-free completed primary boundary-ray factorization $\mathfrak{D}_0^{r,s}$, computable order by order; its descendent extension $\mathfrak{D}^{r,s}$ is interpreted coefficientwise on slope cuts. We write θ_ρ for the factor supported on the ray ρ . For $x^5 + y^5$ its walls begin*

$$\begin{aligned}\theta_{(1,2)} &= \exp\left(\frac{1}{25} t_{2,3} t_{3,3} X_{0,1} + \cdots\right), & \theta_{(2,1)} &= \exp\left(\frac{1}{25} t_{3,2} t_{3,3} X_{1,0} + \cdots\right), \\ \theta_{(1,1)} &= \exp\left(\frac{1}{50} t_{3,3}^2 X_{1,1} + \cdots\right).\end{aligned}$$

The subscripts label primitive boundary directions. A wall factor collects all critical-graph terms for which (k_1+1, k_2+1) lies on that ray (Proposition 2.2).

Theorem 1.4 (Effective determination; §4). *On each $(J, \mathbf{d})$ -diagonal the wall directions span, in the NS-visible case (both residues in the Neveu–Schwarz range: $r(J) \leq r-2$ and $s(J) \leq s-2$), exactly the kernel of the good-basis period functional, and the chamber-index conditions of [GKT22] amount to one Γ /period-weighted linear condition per diagonal with right-hand side determined by strictly lower order. Together with the characterization $\Omega_{r,s}(I) = \Upsilon_{r,s}(I)$ of [GKT22, Cor. 5.14] ($\Upsilon_{r,s}(I)$ the set of solutions of the chamber-index conditions) this determines all open FJRW invariants of $x^r + y^s$, up to the torsor action, from the normalizations and the closed extended theory: the topological recursion of [GKT22, Thm. 0.5] appears as the direction transverse to the walls.*

Theorem 1.4 says that the boundary-ray action, together with the period functional, determines the open theory up to that action. The worked case $x^5 + y^5$ (§5, §6) produces, beyond the wall functions above, the invariant relations and values

$$(2) \quad \langle \tau_0^{(3,3)} \tau_0^{(3,3)} \sigma_1 \sigma_2 \sigma_{12} \rangle + \langle \tau_0^{(3,3)} \tau_0^{(3,3)} \sigma_1^6 \sigma_2 \sigma_{12} \rangle = -\frac{2}{5},$$

$$(3) \quad \langle \tau_0^{(3,3)} \tau_0^{(3,3)} \tau_0^{(3,3)} \sigma_1^4 \sigma_2^4 \sigma_{12} \rangle = \frac{1}{5}, \quad \langle \tau_0^{(1,2)} \tau_0^{(2,1)} \tau_0^{(2,2)} \sigma_{12} \rangle = -\frac{1}{25},$$

the first a wall relation (only the sum is invariant; the difference jumps across the first symmetric pro-nilpotent primary wall $t_{3,3}^2 X_{1,1}$), the second and third absolute: they live on diagonals with no walls, the second on a diagonal invisible to the good-basis periods. The seed relation (for the seed system of Theorem 1.5 below) takes the uniform form

$$(4) \quad \frac{\nu_{(r,0)}}{r} + \frac{\nu_{(0,s)}}{s} = -\frac{1}{rs} \quad (\text{e.g. } \langle \sigma_1^r \sigma_{12} \cdots \rangle + \langle \sigma_2^r \sigma_{12} \cdots \rangle = -\frac{1}{r} \text{ at } r = s),$$

for every two-insertion seed diagonal $\{(a_1, b_1), (a_2, b_2)\}$, $|J| = 2$, with $a_1 + a_2 = r$, $b_1 + b_2 = s$ (Remark 4.3). Here $\nu_{(k_1, k_2)}$ denotes the invariant with boundary profile $\sigma_1^{k_1} \sigma_2^{k_2} \sigma_{12}$, and the suppressed insertions are the two interior markings of the seed.

The last three results are geometric: the transport representation, the realization of each finite quotient by homotopies of boundary conditions, and the exact position-space obstruction carried by that realization.

Theorem 1.5 (The transport representation; §7). *Fix the anchor assignment: each monomial $u_{J, \mathbf{d}} x^{k_1} y^{k_2}$ of the potential is anchored at the one-sided slope cut through its direction (k_1, k_2) , on the side toward the $(1, 1)$ -direction. A constant monomial is anchored at the base cut; on slope 1 either one-sided convention gives the same coefficient. Then, by induction in the divisibility order of deformation monomials: on every monomial $u_{J, \mathbf{d}}$ the difference between the potential's coefficient vector and the transported contributions of the seeds of the proper divisor monomials is independent of the cut, and defining the seed system by these constants, every cut potential is the coefficientwise transport of the seeds,*

$$W(q) = \sum_m \Psi(q) \Psi(q_m)^{-1}(\text{seed}_m),$$

the sum over the monomials m of the potential and q_m the anchor cut of m . On the coefficient of a fixed deformation monomial u , project every factor to the finite divisor-closed quotient consisting of the divisors of u ; only finitely many projected boundary-ray factors are nontrivial, and their slope-ordered product defines $\Psi(q)$ coefficientwise from the base cut to q . The seed system is the unique one with this property. On a coefficient receiving no foreign bending, the seed equals its anchored value; for a foreign-free extreme coefficient of a wall-bearing diagonal that value is its vanishing extreme-index value (Proposition 7.3), so every such coefficient carries seed zero. Those zero-seed coefficients form the purely tropical part of the theory, while the nonzero seed values are non-tropical input.

Theorem 1.5 is the structural core of the tropical description of the open theory: the invariants are computed by broken lines (monotone transports bending at the walls of $\mathfrak{D}^{r,s}$) from canonical seed data. The seeds play the role of the initial monomials of broken lines in [Gro10, CPS24]; the word carries no cluster-theoretic meaning here. The ordered-chain formula and the explicit zero-cycles below give them a zero-dimensional interpretation in rational oriented zero-bordism. Theorem 1.7 determines the exact boundary of the position-space reading: the geometric continuation exists on the punctured sector but factors through slope and cannot supply origin monodromy. The whole is the Landau–Ginzburg analogue of the tropical disk-counting picture of [Gro10, CPS24].

Theorem 1.6 (Geometric realization in a finite coefficient quotient; §7.3). *Let S be a finite divisor-closed set of deformation monomials containing 1. Choose a finite labeled set I with at least*

$$n_{a,b}(S) := \max_{u \in S} \sum_{d \geq 0} \text{ord}_{t_{a,b,d}}(u)$$

labels of every twist (a, b) occurring in S . The quotient factorization has finitely many rays $\rho_1, \dots, \rho_n$, listed in increasing slope. There is a canonical homotopy of boundary conditions, obtained by concatenating families of simple canonical homotopies in the sense of [GKT22, §3.4], from a family realizing $\nu_I^{\min}$ to one realizing $\nu_I^{\max}$, the restrictions of the universal extreme indices, such that after reduction to the quotient supported on S every nonzero jump is supported on a single boundary ray, the rays carrying nonzero reduced jumps occur in increasing slope as

the homotopy time increases, and the total reduced jump on ρ_i is the wall automorphism θ_{ρ_i} of Theorem 1.3.

The staging is forced: above the central-charge threshold a single normal-form family of [GKT22, Lem. 3.40] provably cannot separate the walls of a diagonal carrying two of them, since they all jump at one time (Proposition 7.21), and the exact first order at which two walls share a diagonal is computed for every hyperbolic pair (Proposition 7.19, Remark 7.24). Realizing each boundary-ray factor in its own stage is what evades the obstruction.

Theorem 1.7 (Projective direction sector and obstruction to a singular integral-affine base; §7.4). *For every finite divisor-closed coefficient set S , the wall-stratified punctured sector is the product of a radial half-line and the interval of directions, and it retracts stratum-preservingly to a constant-radius copy of that interval. After a finite staged realization has been chosen, a tame path induces the product of the factors $\theta_{\rho,S}^{\pm 1}$ in its oriented crossing word. Cancellation of an adjacent inverse pair is realized by a relative two-parameter canonical family, unique up to a relative three-parameter family. The reduced product depends only on angular endpoint chambers; equivalently, this continuation observable factors through*

$$\pi : C_{\partial}^{\times} \longrightarrow \mathbb{P}(C_{\partial}),$$

so every radial path acts trivially and dilation changes no such observable. Moreover $C_{\partial}^{\times} \cong (0, \infty) \times [0, 1]$ is contractible: the product $\prod_{\rho}^{\rightarrow} \theta_{\rho} = g^{\min \rightarrow \max}$ is transport between the two boundary sides, not origin monodromy. Hence the present GKT continuation structure does not by itself provide the radially detected modulus, singular affine extension, or vertex monodromy required of the proposed position space. A Gross–Siebert/SYZ and theta-function theory requires additional valuation, affine-gluing, and binary enumerative input.

1.3. Two apparent discrepancies in the literature. Our computations bear on two apparent discrepancies in [GKT22, GKT26], recorded here and in Remarks 4.3 and 2.3 with the supporting derivations. First, imposing the open mirror theorem [GKT22, Thm. 0.3] directly on the oscillatory expansion of the potential — the mechanism of their Theorem 0.2 — gives the seed values (4); at $r = s = 4$ this yields $-\frac{1}{4}$ for the sum in [GKT26, Ex. 5.5], where the value $-\frac{1}{8}$ is stated. The partition count in the expansion of $\mathcal{A}$ forced by [GKT22, Notation 3.42 and Thm. 5.1] (ordered partitions of the labeled marking set, with $1/h!$) also reproduces, through the proven [GKT22, Thm. 5.4], the closed value $\mathcal{A}(\{(1, 1), (1, 1)\}, 0) = \langle \tau_0^{(1,1)3} \rangle = 1$ at $r = s = 5$, pinning the same count independently. This apparent factor-of-two discrepancy is localized to the numerical evaluation in [GKT26, Ex. 5.5], not to the open mirror theorem itself. Second, at wall-bearing marginal monomials the critical graphs have torus bidegree $(0, 0)$, whereas the pro-nilpotent algebra of [GKT22, Def. 4.23] excludes that bidegree. The proof of [GKT22, Thm. 5.9] appears to require the marginal summands: its Q_0 -sets impose the dimension constraint, which admits $(0, 0)$. The realization argument in the proof of [GKT22, Thm. 5.13], which underlies [GKT22, Cor. 5.14], uses Lemma 5.17 to supply the corresponding marginal adjustment. We therefore formulate all statements for the group enlarged by the marginal torus summands, the elliptic wall-crossing group appearing in Maher’s thesis [Mah24, §5.2.2].

1.4. Method. The argument combines the grading and torsor action of the GKT wall-crossing group with ordered factorization in a strictly convex cone. The period relations determine the transverse coordinate on each family of balanced coefficients, while the GKT extension lemma realizes each prescribed finite sequence of group factors by canonical homotopies. The calculations for $x^5 + y^5$ use exact polynomial and rational arithmetic and compare the open

topological recursion with closed correlators obtained independently from Saito flat coordinates, tautological intersections, and Chiodo's formula (§6). Section 8 records the finite checks; the replication materials are available with the paper at <https://github.com/bwuebben/open-fjrw-scattering>. The central-charge threshold and first-wall classification are proved in the companion note [Wue26].

2. THE WALL-CROSSING ALGEBRA OF CRITICAL GRAPHS

We recall the objects of [GKT22, §2.4, §4.3] in the notation used throughout. Internal markings carry twists (a, b) , $0 \leq a \leq r-2$, $0 \leq b \leq s-2$, and descendent exponents $d \geq 0$; boundary markings are of three types, σ_1 (twist $(r-2, 0)$), σ_2 (twist $(0, s-2)$), and the fully twisted σ_{12} (twist $(r-2, s-2)$). For a multiset J of internal twists with descendents $\mathbf{d}$, set $r(J) \equiv \sum_j a_j \pmod{r}$, $s(J) \equiv \sum_j b_j \pmod{s}$, and $m(J, \mathbf{d})$ as in (1).

Definition 2.1 ([GKT22, Not. 2.40]). Write $\sum_j a_j = r(J) + \ell_1 r$, $\sum_j b_j = s(J) + \ell_2 s$ and $N = N(J, \mathbf{d}) = \ell_1 + \ell_2 - |J| + 1 + \sum_j d_j$. The *balanced* disk types over $(J, \mathbf{d})$ are $\Gamma_{J,p}$, $0 \leq p \leq N$, with boundary degrees $(k_1, k_2) = (r(J) + pr, s(J) + (N-p)s)$ and one σ_{12} -point; the *critical* graphs are $\Lambda_{J,p}$, $1 \leq p \leq N$, with boundary degrees $(r(J) + (p-1)r + 1, s(J) + (N-p)s + 1)$ and no σ_{12} -point. We refer to the set of balanced coefficients over a fixed $(J, \mathbf{d})$ as its *diagonal*. The balanced range is empty for $N < 0$, and the critical range is empty for $N < 1$. By [GKT22, Prop. 2.31], the diagonal is the set of lattice points of $sk_1 + rk_2 = m(J, \mathbf{d})$ obeying the congruences.

Proposition 2.2 (The primitive-form shift). *The Lie bidegree of the critical graph $\Lambda_{J,p}$, in the sense of (1), is its boundary degree minus $(1, 1)$ ([GKT22, Prop. 2.39]); equivalently, the wall element attached to Λ is the Hamiltonian vector field, for the primitive form $dx \wedge dy$, of the boundary monomial $x^{\#\sigma_1} y^{\#\sigma_2}$ of the critical disk. With $g_m = z^m(m_2 x \partial_x - m_1 y \partial_y)$ the tropical-vertex generator (Hamiltonian for $d \log x \wedge d \log y$), one has $X_{k_1, k_2} = (xy)^{-1} g_{(k_1+1, k_2+1)}$: the algebra is graded by an integral-affine lattice, the origin shifted by the primitive form, and the wall-crossings are the rational-power tropical vertex maps of [GKT22, Lem. 4.21]. The shift is the removal of the σ_{12} -insertion.*

Proof. The first statement is [GKT22, Prop. 2.39(b)] read against Definition 2.1. The dictionary $X_H^{\text{tor}} = xy X_H^{\text{aff}}$ follows from $\omega_{\text{aff}} = xy \omega_{\text{tor}}$, and the explicit wall maps are [GKT22, Lem. 4.21]. $\square$

Proof of Theorem 1.1. The constraint (1) reads $sk_1 + rk_2 = w(J, \mathbf{d})$ with $k_1, k_2 \geq 0$: for $w < 0$ there are no solutions, for $w = 0$ only $(0, 0)$, and for $w > 0$ the N solutions of Definition 2.1. A twist (a, b) contributes $sa + rb - rs \leq s(r-2) + r(s-2) - rs = rs(\hat{c}_W - 1)$ to the weight, with equality at the maximal twist; descendents contribute $rs d_j > 0$. Hence positive-weight $d = 0$ monomials exist if and only if $\hat{c}_W > 1$, and for $\hat{c}_W = 1$ every $d = 0$ monomial has $w \leq 0$ with equality exactly on products of marginal twists; whenever such an equality monomial is wall-bearing, its unique direction is the torus one. For the existence of walls (not merely of positive-weight monomials) when $\hat{c}_W > 1$: take m interior markings of maximal twist $(r-2, s-2)$; then $N = m - \lceil 2m/r \rceil - \lceil 2m/s \rceil + 1 \geq 1$ once $m(1 - \frac{2}{r} - \frac{2}{s}) \geq 2$, while the weight is $mrs(\hat{c}_W - 1) > 0$, so the resulting critical graphs carry wall summands; this is the maximal-twist construction of [Wue26]. At $\hat{c}_W = 1$, those analogous multisets for which $N \geq 1$ are critical and marginal, realizing the torus wall-crossing; for $\hat{c}_W < 1$, $\ell_1 + \ell_2 \leq m\hat{c}_W < m$ forces $N \leq 0$: no primary critical graphs at all. The qualifier in the theorem is necessary: acting on a lower divisor coefficient multiplies deformation monomials, so nonlinear transport can contribute to a nonpositive-weight target even though no generator is supported there. $\square$

Remark 2.3 (The marginal torus direction). At a wall-bearing marginal monomial ($w = 0$ and $N \geq 1$; necessarily $r(J) = s(J) = 0$ and $N = 1$) the critical graph exists (for $x^4 + y^4$ it is precisely the disk of [GKT26, Fig. 9, Ex. 4.9]), but its Lie bidegree is $(0, 0)$, the direction $x\partial_x - y\partial_y$ excluded from $\mathfrak{g}^{r,s}$ in [GKT22, (4.16),(4.19)] for nilpotency of the (x, y) -filtration. The exclusion is an artifact: with coefficients in the ideal generated by the deformation variables, the exponential converges adically, and the proofs of [GKT22, Thm. 5.9] (whose jump sets $Q_0(A, \mathbf{d})$ impose only (1), admitting $(0, 0)$) and of [GKT22, Cor. 5.14] (whose realizability step moves the two adjacent balanced coefficients by $h \cdot ((-1)^{k_2} s k_1(\Lambda), (-1)^{k_2+1} r k_2(\Lambda))$, at a marginal diagonal exactly the antisymmetric torus motion) use the torus summands. We therefore work with the enlarged algebra

$$\tilde{\mathfrak{g}}^{r,s} = \mathfrak{g}^{r,s} \oplus \bigoplus_{\substack{\emptyset \neq J, w(J, \mathbf{d})=0, N(J, \mathbf{d}) \geq 1 \\ r(J)=s(J)=0}} \mathbb{Q} u_{J, \mathbf{d}} (x\partial_x - y\partial_y),$$

whose new summands are abelian among themselves and act diagonally on the graded pieces. This is a Lie algebra: for $D = X_{0,0}$, $[uD, vX_{a,b}] = (a-b)uvX_{a,b}$ and $[uD, vD] = 0$; since a marginal coefficient u has weight and residues zero, uv satisfies the same critical-generator constraints as v . In the primary algebra nothing is added when $\hat{c}_W < 1$; for the parabolic pairs the primary group is purely torus, matching the elliptic wall-crossing group of Maher [Mah24, §5.2.2]; for $\hat{c}_W > 1$ the torus summands sit over the marginal submonomials. All statements below are over $\tilde{\mathfrak{g}}^{r,s}$ and its group $\tilde{G}^{r,s} = \exp(\tilde{\mathfrak{g}}^{r,s})$; the next lemma supplies the torsor properties over the enlargement.

Lemma 2.4 (All-residue Γ -functionals). *For every nonempty labeled marking set J and descendent vector $\mathbf{d}$, the polynomial $\mathcal{A}(J, \mathbf{d}, \nu)$ is preserved by every generator of $\tilde{\mathfrak{g}}^{r,s}$, including when $r(J) = r - 1$ or $s(J) = s - 1$.*

Proof. Take $\hbar < 0$ and expand, coefficientwise in the deformation variables, the positive-quadrant Laplace integral

$$\mathcal{L}_{\hbar}(W) = \int_0^\infty \int_0^\infty e^{W/\hbar} dx dy.$$

The monomial moments are

$$\int_0^\infty x^q e^{x^r/\hbar} dx = \frac{1}{r} (-\hbar)^{(q+1)/r} \Gamma\left(\frac{q+1}{r}\right),$$

and similarly for y . Therefore the coefficient of $u_{J, \mathbf{d}}$ in $\mathcal{L}_{\hbar}(W^\nu)$ is a fixed nonzero power of $-\hbar$, multiplied by fixed nonzero Gamma factors depending only on $r(J), s(J)$, times $\mathcal{A}(J, \mathbf{d}, \nu)$. This is the termwise calculation in the proof of [GKT22, Thm. 4.9], before the good-basis Kronecker factors are imposed. Unlike those good-basis periods, the displayed Laplace functional is nonzero also for residues $r - 1$ and $s - 1$.

Let $v = uX_{a,b}$ be any enlarged generator. The vector field $X_{a,b}$ is divergence-free and tangent to both coordinate axes. Hence, coefficientwise,

$$\left. \frac{d}{dt} \right|_{t=0} \mathcal{L}_{\hbar}(e^{tv} W) = \int_{[0, \infty)^2} v(e^{W/\hbar}) dx dy = 0 :$$

integration by parts has no contribution on the axes, and the exponential decay removes the boundary at infinity. The same identity after replacing W by $e^{tv} W$ shows that the full exponential action preserves $\mathcal{L}_{\hbar}$, and comparison of deformation coefficients proves the claim. Equivalently, the calculation is the Gamma recurrence $\Gamma(z+1) = z\Gamma(z)$ on every residue class. $\square$

Lemma 2.5 (The enlarged torsor). *The action of $\tilde{G}^{r,s}$ on potentials preserves $\Omega_{r,s}(I)$ and is free and transitive on it, so that $\Omega_{r,s}(I)$ is a torsor in the strict sense, and the jump automorphisms of [GKT22, Thm. 5.9] lie in $\tilde{G}^{r,s}$.*

Proof. Membership of the jumps: the jump sets $Q_0(A, \mathbf{d})$ in the proof of [GKT22, Thm. 5.9] impose only (1), so the displayed jump elements are sums of pro-nilpotent and marginal torus summands.

Preservation: the coefficient of every generator has positive deformation degree, so its exponential converges in that filtration. Lemma 2.4 preserves all the $\mathcal{A}$ -conditions of [GKT22, Def. 3.47]. It also preserves the normalization there: a primary critical monomial has at least two markings, while a descendent singleton does not alter the $\mathbf{d} = \mathbf{0}$ normalization. Thus the enlarged group preserves $\Upsilon_{r,s}(I)$.

Transitivity: first work modulo a finite divisor-closed set of deformation monomials and order those monomials by any linear extension of proper divisibility. Suppose two chamber indices have already been made equal on every proper divisor of $u = u_{J,\mathbf{d}}$. On the diagonal of u , all $h \geq 2$ terms of $\mathcal{A}(J, \mathbf{d}, -)$ involve proper divisors and hence already agree. The difference of the two coefficient vectors therefore lies in the kernel of the nonzero $h = 1$ Gamma functional. If the diagonal has $N + 1$ balanced coefficients, that kernel has dimension N . The N adjacent wall vectors

$$w_p = a_p e_p - b_p e_{p-1}, \quad a_p, b_p > 0, \quad 1 \leq p \leq N,$$

are independent and lie in the kernel by Lemma 2.4; hence they span it. On a marginal diagonal $N = 1$, and the sole vector is the adjoined torus vector $re_1 - se_0 = X_{0,0}(W_0)$. A unique combination of generators with coefficient u therefore removes the difference. Its action on every nonconstant term of the potential carries a strict multiple of u , so it does not disturb the completed divisibility step. Induction proves transitivity in every finite quotient, and the compatible corrections pass to the coefficientwise completion.

The equality $\Omega_{r,s}(I) = \Upsilon_{r,s}(I)$ remains the realization theorem [GKT22, Thm. 5.13 and Cor. 5.14]. Its inclusion $\Upsilon_{r,s}(I) \subseteq \Omega_{r,s}(I)$ does not require the omitted marginal transitivity: at equation (5.50) of [GKT22], the two required $\mathcal{A}$ -values agree directly because both are the prescribed value of Definition 3.47. The enlarged action therefore preserves and is transitive on the realizable indices.

Freeness: let $g = \exp(v)$ with $v \neq 0$, and let u be minimal in the divisibility order among the monomials in the support of v . The change of the u -diagonal of any potential is exactly $v_u(W_0)$, $W_0 = x^r + y^s$: contributions of other summands, or of higher exponential terms, carry monomials properly containing a support monomial, which cannot equal u by minimality. And $v_u(W_0) \neq 0$. Indeed, on the diagonal belonging to u , the N pro-nilpotent generators act by the adjacent vectors $a_p e_p - b_p e_{p-1}$ with $a_p, b_p > 0$. These vectors are linearly independent: reading a relation successively in the $e_N, e_{N-1}, \dots, e_1$ coordinates kills all its coefficients. On a marginal diagonal there is instead the single nonzero torus vector $re_1 - se_0$. Thus no nonzero linear combination of the summands in v_u can cancel. (Individually, a pro-nilpotent summand contributes $r(k_2+1)x^{k_1+r}y^{k_2} - s(k_1+1)x^{k_1}y^{k_2+s}$, while a torus summand contributes $rx^r - sy^s$.) So g moves every potential, which is freeness and not merely faithfulness. The distinction matters downstream: uniqueness of the transition element, used in Theorem 1.2(i), needs trivial stabilizers, and faithfulness alone does not supply them for a nonabelian group. The argument of [GKT22, Thm. 4.27, Step 2] likewise establishes freeness, though it is stated there as faithfulness. All uses below are through this lemma. $\square$

3. SLOPE-ORDERED BOUNDARY-RAY FACTORIZATION

Throughout, a *normal-form family* is a family of simple canonical homotopies supplied by [GKT22, Lem. 3.40]: for each subset A of the internal labels there is one time $t_A \in (0, 1)$, all times distinct, and every jump of a critical graph with internal marking set A occurs at t_A . The times are auxiliary choices (Remark 7.20).

Proof of Theorem 1.2(i). Let $(\mathfrak{s}_t)_{t \in [0,1]}$ be a family of simple canonical homotopies supplied by [GKT22, Lem. 3.40], with $\mathfrak{s}_1 = \mathfrak{s}_0$ and jump times $t_1 < \dots < t_M$ and jump elements $g_i \in \tilde{G}^{r,s}$, each a sum over the critical graphs with internal markings A_i , with coefficients, up to the explicit signs $(-1)^{|A_i|+k_2}$ (k_2 the Lie bidegree) of the display of g_i there, the zero counts $\Delta(\Lambda)$ of the homotopy on the critical stratum ([GKT22, proof of Thm. 5.9, eqns. (5.33)–(5.37)]). By that proof the potential $W^{\mathfrak{s}_t}$ is constant between jumps and transforms by g_i across t_i ; around the loop, $W^{\mathfrak{s}_0} = (g_M \circ \dots \circ g_1)(W^{\mathfrak{s}_0})$, and freeness (Lemma 2.5) gives $g_M \circ \dots \circ g_1 = \text{id}$. For two such paths with fixed endpoints, their total products g, g' satisfy $g'g^{-1}(W^{\mathfrak{s}_1}) = W^{\mathfrak{s}_1}$, hence $g' = g$. $\square$

Proof of Theorem 1.2(ii). Let $\mathfrak{m}$ be the deformation ideal and first work in the primary quotient

$$F^p \tilde{\mathfrak{g}}_0^{r,s} := \text{span}\{u_{J,0}X : |J| \geq p\}, \quad L_{<K} := \tilde{\mathfrak{g}}_0^{r,s} / F^K.$$

It is a finite-dimensional nilpotent Lie algebra and $[F^p, F^q] \subseteq F^{p+q}$ (here and below $F^\bullet$ also denotes the induced filtration on the quotient). For a rational boundary ray ρ let L_ρ be the span of the generators uX_{k_1,k_2} for which $(k_1+1, k_2+1) \in \rho$, placing a marginal generator $u(x\partial_x - y\partial_y)$ in $\rho = \mathbb{R}_{\geq 0}(1, 1)$. Then $L_{<K} = \bigoplus_\rho L_\rho$, with only finitely many summands.

Each L_ρ is abelian. Indeed,

$$[X_{k_1,k_2}, X_{k'_1,k'_2}] = \det((k'_1+1, k'_2+1), (k_1+1, k_2+1)) X_{k_1+k'_1, k_2+k'_2},$$

so the bracket vanishes when the shifted directions are parallel; on the $(1, 1)$ -ray the marginal vector field also commutes with every $X_{q,q}$. The deformation coefficient of a marginal generator has positive $\mathfrak{m}$ -degree, so it belongs to the same nilpotent filtration.

Fix any slope order. The multiplication map

$$\prod_{\rho} \exp(L_\rho) \longrightarrow \exp(L_{<K})$$

is bijective. Modulo F^2 this is the direct-sum decomposition above. Suppose the factors have been found through F^n . The logarithm of the residual in F^n/F^{n+1} decomposes uniquely by ρ ; insert these components into the corresponding factors. Commuting a degree- n correction past a previously chosen positive-degree term changes the product only in F^{n+1} . This proves existence by induction. At the first filtration degree where two ordered factorizations differ, the same direct-sum decomposition forces all differences to vanish, proving uniqueness. The construction is compatible as K varies.

For descendents, perform the same construction after quotienting to any finite divisor-closed set of deformation monomials. Its complement is upward closed, so this is a Lie-algebra quotient, and the compatible finite constructions give the asserted coefficientwise presentation. $\square$

Remark 3.1. [GKT22, Thm. 5.9] produces the total jump in a homotopy-time-ordered factorization, one factor per subset of internal markings, mixing wall directions; Lemma 3.40 of [GKT22] shows the jump times and their order are auxiliary choices. The boundary-ray presentation is a canonical reorganization by the *shifted* directions forced by Proposition 2.2, not by the additive

Lie rays (k_1, k_2) . The affine shift does not preserve rays; this filtered distinction is what makes the canonical boundary-ray factorization of Theorem 1.3 geometrically meaningful.

4. PERIOD RELATIONS FOR EACH FAMILY OF BALANCED COEFFICIENTS

Two structural facts from [GKT22] organize this section. The open topological recursion factorizes through a cohomological-field-theory node splitting: in [GKT22, Lem. 6.4] the distinguished insertion splits off a closed extended invariant and continues as an open piece carrying the socle-dual twist $(a, b) \mapsto (r-2-a, s-2-b)$ of the Jacobi ring $\mathbb{C}[x, y]/(x^{r-1}, y^{s-1})$: the node is the residue pairing. And the Γ -weights in the invariant combinations $\mathcal{A}$ are the periods of $x^r + y^s$: $\int_0^\infty e^{-x^r} x^k dx = \frac{1}{r} \Gamma(\frac{1+k}{r})$, so the weight attached to a boundary profile is the period normalization of the primitive form. The good-basis period moments themselves are explicit:

Lemma 4.1 (Periods). *For the good-basis cycles Ξ_a of x^r ($\int_{\Xi_a} x^{a'} e^{x^r/h} = \delta_{aa'}$, $0 \leq a, a' \leq r-2$), the moments $P_a(p) = \int_{\Xi_a} x^p e^{x^r/h}$ satisfy $P_a(p) = \delta_{ap}$ for $p \leq r-2$, $P_a(r-1) = 0$, and $P_a(p) = -\frac{h(p-r+1)}{r} P_a(p-r)$.*

Proof. Total derivatives integrate to zero on Ξ_a : $0 = \int_{\Xi_a} \frac{d}{dx} (x^q e^{x^r/h}) = q P_a(q-1) + \frac{r}{h} P_a(q+r-1)$. $\square$

Proof of Theorem 1.4. The case $N = 0$. If $N = 0$, there is no wall direction and the nonzero $h = 1$ weight in $\mathcal{A}$ determines the sole coefficient modulo lower monomials, so the claim is immediate.

Wall-bearing diagonals. Fix now a wall-bearing diagonal $(J, \mathbf{d})$, of positive or marginal weight, with coefficients ν_p , $0 \leq p \leq N$, as in Definition 2.1. The wall element of $\Lambda_{J,p}$ acts on $W_0 = x^r + y^s$ by

$$(5) \quad u_{J,\mathbf{d}} X_{k_1,k_2}(W_0) = u_{J,\mathbf{d}} (r(k_2+1) x^{k_1+r} y^{k_2} - s(k_1+1) x^{k_1} y^{k_2+s}),$$

which lands on the two adjacent balanced points of the same diagonal: in the basis $e_0, \dots, e_N$ of coefficients, the wall p moves along the jump vector $w_p = a_p e_p - b_p e_{p-1}$ (not to be confused with the scalar Euler weight $w(J, \mathbf{d})$) with $a_p = r k_2(\Lambda_{J,p}) > 0$, $b_p = s k_1(\Lambda_{J,p}) > 0$; here and below $k_i(\Lambda)$ denotes the *boundary* degrees of Λ , which exceed the Lie bidegree (k_1, k_2) of the display by $(1, 1)$, so that $r(k_2+1) = r k_2(\Lambda)$ and $s(k_1+1) = s k_1(\Lambda)$, both positive. (These are exactly the geometric jump coefficients of [GKT22, (5.52)–(5.54)].) The w_p are triangular, hence independent, and each is annihilated by the period functional: by Lemma 4.1, $P_a(k_1+r) P_b(k_2) = -\frac{h(k_1+1)}{r} P_a(k_1) P_b(k_2)$ cancels against the second term of (5). *NS-visible diagonals.* On an NS-visible diagonal ($r(J) \leq r-2$ and $s(J) \leq s-2$) the period functional is nonzero on each e_p , so its kernel has dimension N and coincides with $\text{span}(w_p)$. By [GKT22, Thm. 4.9] the coefficient of $u_{J,\mathbf{d}}$ in $\int_{\Xi_{(r(J), s(J))}} e^{W^\nu/h} dx \wedge dy$ equals $(-1)^{|J|} (-h)^{-d(J,\mathbf{d})-2} \mathcal{A}(J, \mathbf{d}, \nu)$, whose $h = 1$ part (h the number of parts in the ordered-partition expansion of $\mathcal{A}$, [GKT22, Notation 3.42 and Thm. 4.9]) is the period-weighted sum $\sum_p \omega_p \nu_p$ over the diagonal and whose $h \geq 2$ parts involve only strictly smaller monomials. The chamber-index conditions [GKT22, Def. 3.47] prescribe the value of $\mathcal{A}$.

Invisible diagonals. On invisible diagonals ($r(J) = r-1$ or $s(J) = s-1$) the good-basis periods vanish identically on the diagonal (Lemma 4.1), but the $h = 1$ weights of $\mathcal{A}$ remain nonzero there (their arguments $(1+K)/r$ are positive integers, where Γ does not vanish), and $\text{span}(w_p)$ lies in the kernel of the weighted functional modulo strictly smaller monomials, by the boundary-condition independence of $\mathcal{A}$ [GKT22, Thm. 5.4]; so the same one-linear-condition structure holds, imposed directly by the conditions, which by [GKT22, Cor. 5.14] characterize

$\Omega_{r,s}(I)$ completely. Ordering monomials by degree and descendents determines every invariant, up to the span of the w_p (the torsor directions), proving the theorem. The recursion [GKT22, Thm. 0.5] is the same system of conditions organized by descendent reduction, whence the last statement. $\square$

Remark 4.2 (A first one-parameter family). For $x^5 + y^5$ the conditions produce, order by order over the $t_{3,3}$ -monomials, the one-insertion normalization $\langle \tau_0^{(3,3)} \sigma_1^3 \sigma_2^3 \sigma_{12} \rangle = 1$; the first-wall relation (2) at $t_{3,3}^2$; the absolute value $\frac{1}{5}$ of (3) at $t_{3,3}^3$ (a diagonal with $N = 0$, hence no walls, and $r(J) = 4 = r-1$, hence invisible to the good-basis periods, so the condition that determines it is not a period condition); and at $t_{3,3}^4$ a relation quadratic in the first-wall invariants, giving $\nu_{(2,7)}^{\min} = -\frac{12}{25}$ in the normal form of Theorem 1.3. The descendent seed at $d = 1$ gives $\langle \tau_1^{(3,3)} \sigma_1^8 \sigma_2^3 \sigma_{12} \rangle + \langle \tau_1^{(3,3)} \sigma_1^3 \sigma_2^8 \sigma_{12} \rangle = -\frac{5}{4}$.

Remark 4.3 (The seed values, and [GKT26, Ex. 5.5]). Imposing [GKT22, Thm. 0.3] on the oscillatory expansion of the potential directly — the mechanism by which [GKT22, Thm. 0.2] is proved, so that no combinatorial convention enters — forces, at every two-insertion seed diagonal $\{(a_1, b_1), (a_2, b_2)\}$ with $a_1 + a_2 = r$, $b_1 + b_2 = s$, $\mathbf{d} = 0$, the relation (4): the diagonal supports the boundary profiles $(r, 0)$ and $(0, s)$ only, $d(J, 0) < 0$ so the mirror theorem admits no term there, and the linear and quadratic insertions contribute $\nu_{(r,0)}/r + \nu_{(0,s)}/s$ and $1/(rs)$ respectively (Lemma 4.1). At $r = s = 4$ this gives $-\frac{1}{4}$ for the sum considered in [GKT26, Ex. 5.5], where $-\frac{1}{8}$ is stated; the discrepancy traces to the $h = 2$ partition count in the evaluation of $\mathcal{A}$ there, which [GKT22, Notation 3.42 and Thm. 4.9] fixes as the count of ordered partitions of the labeled marking set with $1/h!$. The same count is pinned independently by the proven [GKT22, Thm. 5.4]: at $r = s = 5$, $\mathcal{A}(\{(1, 1), (1, 1)\}, 0)$ equals the closed three-point value $\langle \tau_0^{(1,1)3} \rangle = 1$, which the labeled count reproduces and the halved count does not (§8). On the marginal seed diagonals the two coefficients in (4) are moved (antisymmetrically) only by the torus direction of Remark 2.3; the symmetric gauge value is $\nu_{(r,0)} = \nu_{(0,s)} = -\frac{1}{2r}$ at $r = s$. The discrepancy concerns only the worked example; no theorem of [GKT22] or [GKT26] is affected.

5. EXTREMAL NORMAL FORMS AND THE BOUNDARY-RAY FACTORIZATION $\mathfrak{D}^{r,s}$

Proof of Theorem 1.3. Work in a primary quotient modulo F^K , or more generally on a finite divisor-closed coefficient set as in Theorem 1.2(ii). For existence and uniqueness of $\nu^{\min}$, order its monomials by total degree. At each step, on each wall-bearing diagonal the span of the wall vectors $w_p = a_p e_p - b_p e_{p-1}$ ($a_p, b_p > 0$) is complementary to $\mathbb{Q}e_0$: a combination $\sum c_p w_p \in \mathbb{Q}e_0$ forces, reading the $e_N, e_{N-1}, \dots$ components in turn, $c_N = \dots = c_1 = 0$. Hence there is a unique group correction at this order making all ν_p , $p \geq 1$, vanish, and it does not disturb lower orders; on marginal diagonals the torus direction plays the role of the single wall. The resulting index is enumerative because the group preserves $\Omega_{r,s}(I)$ (Lemma 2.5). Uniqueness follows from freeness (Lemma 2.5) plus the same triangularity: a nontrivial leading correction moves some $\nu_{p \geq 1}$. The construction for $\nu^{\max}$ is symmetric (complement of $\mathbb{Q}e_N$). The boundary-slope-ordered factorization of the unique $g^{\min \rightarrow \max}$ (Theorem 1.2(ii)) is then canonical, and its wall functions are computed order by order from the normal forms. Uniqueness makes these constructions compatible under restriction, giving the primary completion and the coefficientwise descendent extension. $\square$

Example 5.1 (The first walls of $\mathfrak{D}^{5,5}$). At order two in the deformation variables the three wall-carrying diagonals of $x^5 + y^5$ have balanced profiles $(0, 6)$, $(5, 1)$; $(6, 0)$, $(1, 5)$; and $(1, 6)$, $(6, 1)$

respectively, with invariant relations and normal forms (§8):

monomial	relation	normal forms
$t_{2,3}t_{3,3}$	$\frac{2}{5}\nu_{(0,6)} + \frac{1}{5}\nu_{(5,1)} = -\frac{2}{25}$	$\nu_{(0,6)}^{\min} = -\frac{1}{5}, \nu_{(5,1)}^{\max} = -\frac{2}{5}$
$t_{3,2}t_{3,3}$	the image under $x \leftrightarrow y$	by symmetry
$t_{3,3}^2$	$\frac{2}{5}(\nu_{(1,6)} + \nu_{(6,1)}) = -\frac{4}{25}$	$\nu_{(1,6)}^{\min} = \nu_{(6,1)}^{\max} = -\frac{2}{5}$

Solving $W^{\nu^{\max}} - W^{\nu^{\min}} = c \cdot u X_{k_1, k_2}(W_0)$ on each diagonal is an overdetermined system (two monomial components, one coefficient); the relations above force its consistency and yield the ray factors of Theorem 1.3. The wall $t_{3,3}^2 X_{1,1}$ is the first symmetric pro-nilpotent primary wall: only the sum (2) is invariant, the difference jumps.

6. CLOSED FJRW CORRELATORS FOR $x^5 + y^5$ AND THE OPEN TOPOLOGICAL RECURSION

The determination of Theorem 1.4 consumes closed extended FJRW invariants. For the first symmetric hyperbolic Fermat pair $x^5 + y^5$ ($\hat{c}_W = \frac{6}{5} > 1$) we compute this input exactly.

Example 6.1 (Closed vertex weights for $x^5 + y^5$). By Sebastiani–Thom the state space and cohomological field theory of the Fermat sum factor as $A_4(x) \otimes A_4(y)$, the tensor square of the 5-spin (A_4) theory; a closed invariant $\langle \prod_i \tau_{d_i}^{(a_i, b_i)} \rangle$ is the tensor correlator $\int_{\overline{\mathcal{M}}_{0,n}} W^x(\mathbf{a}) W^y(\mathbf{b}) \psi^{\mathbf{d}}$ with W^x, W^y the two Witten classes, of degrees $\varepsilon_1 = (\sum a_i - 3)/5$ and $\varepsilon_2 = (\sum b_i - 3)/5$. We compute these exactly (primitive-form normalization, sign $(-1)^{\varepsilon_1 + \varepsilon_2}$) by three independent, mutually consistent routes:

- the single-variable A_4 invariants from the Saito primitive form (flat coordinates, three-point residues, string/dilaton/topological-recursion descendents), validated against the A_2 normalization $\langle (\tau_0^1)^4 \rangle = \frac{1}{3}$, WDVV for A_2, A_3, A_4 , and the $r = 2$ Witten–Kontsevich numbers;
- the factorizing tensor correlators ($\varepsilon_1 = 0$ or $\varepsilon_2 = 0$), where one Witten class is the topological unit;
- the fully mixed correlators with $\min(\varepsilon_1, \varepsilon_2) = 1$, by reconstructing the divisor $\mathcal{W} = \sum_U c_U \delta_U$ on $\overline{\mathcal{M}}_{0,n}$ from its node-splitting restrictions and pairing with the other class, using a self-contained genus-zero tautological intersection calculus (splitting recursion, $\delta_S^2 = -\psi' - \psi''$), verified against Witten–Kontsevich and the Petersen intersection form of $\overline{\mathcal{M}}_{0,5} = \text{dP}_5$; each reconstructed class passes the independent check $\langle \mathcal{W} \psi^{n-4} \rangle = \langle \tau_1^{a_i} \cdots \rangle_{A_4}$.

Representative values (denominators 5^{n-3} ; the socle pairing is $\langle \tau_0^{(0,0)} \tau_0^{(0,0)} \tau_0^{(3,3)} \rangle = 1$):

$$\begin{aligned}
\langle (\tau_0^{(2,1)})^3 \tau_0^{(2,0)} \rangle &= -\frac{2}{5}, & \langle \tau_0^{(1,3)} \tau_0^{(1,0)} (\tau_0^{(3,0)})^2 \rangle &= -\frac{1}{5} & (\varepsilon_1, \varepsilon_2) &= (1, 0), \\
\langle (\tau_0^{(2,1)})^2 \tau_0^{(2,2)} (\tau_0^{(1,2)})^2 \rangle &= \frac{4}{25}, & \langle (\tau_0^{(2,2)})^3 (\tau_0^{(1,1)})^2 \rangle &= \frac{2}{25} & (1, 1), \\
\langle (\tau_0^{(2,3)})^2 \tau_0^{(1,3)} \tau_0^{(1,2)} (\tau_0^{(1,1)})^2 \rangle &= -\frac{4}{125}, & \langle (\tau_0^{(2,3)})^2 (\tau_0^{(1,3)})^3 \tau_0^{(1,2)} \tau_0^{(0,1)} \rangle &= \frac{6}{625} & (1, 2), (1, 3).
\end{aligned}$$

The smallest case with both classes of codimension two, $n = 7$, is computed by the codimension-two stratum intersection form: e.g. $\langle \tau_0^{(3,1)} \tau_0^{(2,1)} (\tau_0^{(2,2)})^3 \tau_0^{(1,2)} \tau_0^{(1,3)} \rangle = \frac{34}{625}$, invariant under relabelling.

For $x^5 + y^5$, substituting the closed extended invariants into the open topological recursion of [GKT22, Thm. 0.5] gives equality in all 72 admissible instances with $|J| = 2$ and all 105 with $|J| = 3$, including the extended (Ramond, $\text{tw} = -1$) sectors. The calculation evaluates Witten’s

r -spin class through Chiodo's formula and independently reproduces the values in Example 6.1, including $\frac{34}{625}$ and the descendent correlators. Section 8 describes the exact checks.

7. BOUNDARY DIRECTIONS, POTENTIAL TRANSPORT, AND GEOMETRIC REALIZATION

The results above define the canonical boundary-ray factorization $\mathfrak{D}^{r,s}$ and show that its action, together with the period relations, determines the invariants. What they do not provide is a *position space*: an intrinsic domain in which the walls are rays and the invariants are computed by broken lines, as for the Fano Landau–Ginzburg models of [Gro10, CPS24]. This section builds as much of that picture as is a theorem and isolates what is not.

Three words of warning on vocabulary. We reserve *base* for the base $B\Gamma$ of a graded graph [GKT22, Def. 3.5], an unrelated notion carrying the inductive structure of the boundary conditions, and call the domain below the *direction sector*; and *consistency* here always means the path-independence of Theorem 1.2(i) in boundary-condition space. The product $g^{\min \rightarrow \max}$ is the transport of a boundary-to-boundary path through the sector, whose two ends carry $\nu^{\min}$ and $\nu^{\max}$; it is not loop holonomy (Theorem 7.26). Likewise *chamber index* is [GKT22]'s name for a system of invariants and does not refer to a chamber of the direction sector.

The order below puts what is proved first. §7.1 places the walls, §7.2 proves the transport representation, its ordered-chain inversion and its rational zero-bordism class, and the sense in which the theory exceeds its own tropicalization; §7.3 realizes every finite coefficient window by staged homotopies and proves the exact obstruction that forces the staging; and §7.4 proves the projective-sector statement and the resulting position-space obstruction.

7.1. The sector and its rays.

Definition 7.1. The *boundary-charge lattice* is $N_\partial = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2$, where e_i records one boundary insertion of type σ_i . The *direction sector* is its effective cone $C_\partial = \mathbb{R}_{\geq 0}e_1 + \mathbb{R}_{\geq 0}e_2$, with the $(1,1)$ -shifted integral-affine structure of Proposition 2.2; its two sides are the rays of the two boundary types. A factor of $\mathfrak{D}^{r,s}$ is assigned to the ray $\mathbb{R}_{\geq 0}(k_1+1, k_2+1)$ determined by its critical graph's boundary monomial, and a chamber of a finite primary truncation is a connected component of the complement of these rays. For the completed descendent factorization, a *slope cut* is a direction together with a choice of side when the direction itself carries a factor. The *base cut* is the side of the sector along the σ_1 -axis ray, and the transport $\Psi(q)$ to a cut q is the slope-ordered product of the ray factors between the base cut and q , defined on each finite divisor-closed coefficient set and then coefficientwise; only the composites $\Psi(q)\Psi(q')^{-1}$ enter below, so the choice of base is a convention. For $u, v \in \mathbb{Z}^2$ write $\text{cross}(u, v) = u_1v_2 - u_2v_1$.

Thus the quadrant, its lattice, and its axes come from the open FJRW boundary data rather than from an auxiliary drawing. Only its projectivization $\mathbb{P}(C_\partial) \cong [0, \infty]$, the ordered slope space, is used by transport. The placement is not a convention but forced, and the same computation shows why the theory looks tropical at all: the $(1,1)$ -shift of Proposition 2.2, which is nothing but the removal of the σ_{12} -insertion, is exactly what makes bending along a transported monomial monotone in slope, hence what makes broken lines behave like broken lines here (Theorem 7.5).

Proposition 7.2 (Forced placement). *The wall element X_{k_1, k_2} fixes exactly the monomials proportional to its boundary-monomial direction (k_1+1, k_2+1) . Hence in any geometric realization the wall of a critical graph must lie on the ray of its boundary monomial (the $(1,1)$ -shifted ray), and a seed monomial is supported on the ray of its own exponent. The direction sector of Definition 7.1 is thus forced, its sides the rays of the two boundary types. If those two sides were artificially identified, their ordered crossing would define a counterfactual loop with holonomy*

$g^{\min \rightarrow \max} \neq \text{id}$. In the actual sector the sides remain distinct, and Theorem 1.7 proves that no such loop exists.

Proof. $X_{k_1, k_2}(x^a y^b) = ((k_2+1)a - (k_1+1)b)x^{a+k_1}y^{b+k_2}$, and the coefficient is the determinant of (a, b) against (k_1+1, k_2+1) , vanishing precisely on the ray. The holonomy statement is Theorem 1.3. $\square$

Proposition 7.3 (Interlacing and extreme support). *On every diagonal the balanced monomials and the walls interlace in slope (writing bal_p for the balanced monomial of $\Gamma_{J,p}$), $\text{bal}_N < \Lambda_N < \dots < \Lambda_1 < \text{bal}_0$, and the normal forms of Theorem 1.3 are the extreme-chamber support conditions in every finite primary truncation: near one axis only the monomial at the far end of each diagonal survives.*

Proof. Immediate from the formulas of Definition 2.1: the wall Λ_p has boundary degree between those of bal_p and bal_{p-1} in slope; the support statement is the normal-form construction read through the interlacing. $\square$

7.2. The transport representation.

Definition 7.4. The *anchor cut* q_m of a potential monomial m is the one-sided cut through its own direction, on the side toward the $(1, 1)$ -direction. If m is off every wall in a finite truncation, this is represented by the chamber containing its direction; for m on the $(1, 1)$ -direction either side gives the same coefficient, and the exponent $(0, 0)$ is anchored at the base cut. Its *anchored value* is its coefficient at q_m . For a candidate assignment of seeds, $T(q) := \sum_m \Psi(q)\Psi(q_m)^{-1}(\text{seed}_m)$ denotes the resulting transport potential.

Every bend along a transported monomial adds $\beta - (1, 1)$ to the exponent, β the boundary direction on the crossed ray. Throughout, seed and target are taken off the crossed rays: a seed on a bend ray has vanishing bend coefficient, and an on-ray target uses the one-sided anchor cut toward the $(1, 1)$ -direction, the convention of Theorem 1.5. Call a bending contribution to a coefficient *foreign* if it originates from a seed other than the coefficient's own.

Theorem 7.5 (Monotonicity of broken-line bends). *Along a path with b bend factors from a seed with exponent m_σ into a monomial m ,*

$$b \text{ cross}(m, (1, 1)) = \text{cross}(m, m_\sigma) + \sum_f \text{cross}(m, \beta_f).$$

Consequently no bending contribution originating from a different initial coefficient reaches a monomial's anchor cut from the far side of the monomial's own direction, the side away from the $(1, 1)$ -direction: a monomial below the $(1, 1)$ -direction receives foreign bending only along paths descending in slope, one above it only along paths ascending, and monomials on the $(1, 1)$ -direction receive no foreign contribution at all.

Proof. Summing the per-bend exponent motion gives $m + b(1, 1) = m_\sigma + \sum_f \beta_f$; the displayed identity is its cross product with m . Transport is monotone, so every crossed ray lies strictly between the seed's direction and m 's. For a path from below, seed direction and all crossed rays are strictly shallower than m , so the right side is negative, forcing $\text{cross}(m, (1, 1)) < 0$, i.e. m strictly above the $(1, 1)$ -direction; from above it is positive, forcing m strictly below. A monomial on the $(1, 1)$ -direction, where the cross product vanishes, excludes both. $\square$

The primitive-form shift is thus precisely the monotone-bending mechanism.

Throughout, *seed* means the initial datum of a broken line in the sense of [Gro10, CPS24] and carries no cluster-theoretic meaning.

Lemma 7.6 (Reduction). *For the anchored-value assignment, $T = W$ at every cut if and only if, for every coefficient, the foreign bending contributions cancel at its anchor cut.*

Proof. Wall automorphisms are additive, so $W - T$ transports as a family; its coefficients on any divisibility-minimal nonzero monomial are cut-independent (all bending contributions come from proper divisors, where $W - T$ vanishes), and evaluating each coefficient at its anchor cut gives zero exactly under the stated cancellation. $\square$

Theorem 7.7 (One-parameter descendent families). *At linear order in a diagonal's own wall functions applied to the axis seeds x^r, y^s , the cancellation of Lemma 7.6 holds on every diagonal of every $(r, s, J, \mathbf{d})$: transporting into the cut at bal_p from below crosses exactly the walls Λ_q , $q \geq p+1$, whose x^r -bends land on bal_q , never on the target coefficient, while from above the crossed walls' y^s -bends land on bal_{q-1} , $q \leq p$, again off the target: the interlacing pattern is the no-backwards-bending property of broken lines in [CPS24]. Bends of divisor seeds at foreign walls are outside this statement; Example 7.15 shows they can arrive. For singleton diagonals (one internal marking, any descendent) no cross-diagonal contribution exists, the linearization is exact, and the transport representation holds with the anchored-value assignment on every such tower, with wall functions the chain $c_{q+1} = c_q a_q / b_{q+1}$ closed by the $\mathcal{A}$ -condition.*

Proof. The crossing sets and landing indices are those of Proposition 7.3 together with the landing identities: by (5), $X_{\Lambda_q}(x^r)$ is a nonzero rational multiple of the monomial bal_q and $X_{\Lambda_q}(y^s)$ of bal_{q-1} . For a singleton monomial no proper divisor exists, so the only contributions are the linear ones. $\square$

Two computed examples below (Examples 7.14 and 7.15) calibrate what the seeds are.

Theorem 7.8 (Transport representation; = Theorem 1.5). *With the anchor assignment of Definition 7.4, the difference between the potential's coefficient vector on a monomial and the transported contributions of the seeds of its proper divisors is independent of the cut. Defining the seed system by these constants, every cut potential is the transport*

$$W(q) = \sum_m \Psi(q) \Psi(q_m)^{-1}(\text{seed}_m)$$

of the seeds, and the seed system is the unique one with this property.

Proof of Theorem 1.5. Induct on the divisibility order of deformation monomials; the base is $W_0 = x^r + y^s$, whose two coefficients are seeds with value 1 in every cut. Fix a monomial $u = u_{J, \mathbf{d}}$ and assume the seeds are defined, with the transport identity holding, on every proper divisor of u . Write D for the diagonal of u , so that $u_D = u$, and E for the diagonal of a proper divisor u_E of u . Let $F(q)$ denote the u -component of the sum of the transported lower seeds.

Step 1: the jump formula. Across any wall $\theta = \exp(\eta)$ the coefficients on D jump by

$$\Delta_\theta(W|_D) = \sum_{j \geq 1} \frac{1}{j!} \sum_{\substack{\eta_1, \dots, \eta_j \in \text{supp}(\eta) \\ u_E u_{\eta_1} \cdots u_{\eta_j} = u_D}} [\eta_1 \cdots \eta_j (W|_E)]_D,$$

a universal expression in the wall function and the coefficients of W on proper divisors E of D and on W_0 : any other source overshoots u , and D 's own walls act on D 's coefficients only through W_0 (their monomial is u itself), so the jump is exactly linear there.

Step 2: cancellation, and chamber-independence of the residual. The jump of F across the same wall is the same universal expression evaluated on the transported-lower-seed sum, whose divisor coefficients equal W 's by the induction hypothesis and whose W_0 -coefficients are the

same constants 1. Hence the two jumps cancel in the difference and $V := W(q)|_D - F(q)$ is independent of q .

Step 3: the seed at D , and uniqueness. A u -term transports with unchanged u -component (every wall action on it overshoots u), so placing $\text{seed}(u) := V$ at the assigned anchor cuts gives $T|_D = V + F = W|_D$ at every cut (all sums are coefficientwise finite: only finitely many divisor seeds and bend factors contribute to a fixed monomial), completing the induction. Uniqueness is immediate: V is determined by W and the lower seeds. Anchor assignments affect only the higher-order terms of the transports, hence the seed values at higher orders, never the existence or uniqueness of the decomposition; the $(1, 1)$ -side assignment is the canonical choice, being the one compatible with Theorem 7.5. $\square$

Lemma 7.9 (Ordered bend expansion). *Work in a finite divisor-closed coefficient quotient and write $z^a = x^{a_1}y^{a_2}$. Suppose a transport path crosses, in chronological order,*

$$\Theta_j = \exp(\epsilon_j \eta_j), \quad \eta_j = \sum_{\lambda \in L_j} c_\lambda u_\lambda X_{k_\lambda}, \quad \epsilon_j \in \{1, -1\}.$$

For nonnegative integers n_j and ordered choices $\lambda_{j,1}, \dots, \lambda_{j,n_j} \in L_j$, read the decorations first by increasing j and put $a^h = a^{h-1} + k_{\lambda_h}$, starting from $a^0 = a$. Then

$$(6) \quad (\Theta_m \circ \dots \circ \Theta_1)(u_E z^a) = \sum_{\gamma} \frac{\prod_h \epsilon_{j(h)} c_{\lambda_h} \det(a^{h-1}, k_{\lambda_h} + (1, 1))}{\prod_j n_j!} u_E \prod_h u_{\lambda_h} z^{a^{|\gamma|}},$$

the finite sum over the surviving decorated bend chains γ .

Proof. Expand each exponential in powers of η_j and distribute each power over its summands. Successive applications use

$$X_k(z^a) = ((k_2 + 1)a_1 - (k_1 + 1)a_2)z^{a+k} = \det(a, k + (1, 1))z^{a+k}.$$

The exponential supplies $1/n_j!$, the path supplies the displayed factor order, and nilpotence makes the sum finite. Generators on one boundary ray commute, so repeated bends may equivalently be grouped as multisets with the same factorials. Notice that the expansion is by chains: transport is linear in its initial monomial. Rooted Lie trees arise only after an additional, noncanonical BCH expansion of the ray factors. $\square$

Index a potential slot by $i = (D, a)$, where D is a deformation monomial and a an exponent on its diagonal; let q_i be its anchor cut. If $j = (E, b)$, define the strictly divisibility-triangular transport kernel

$$(7) \quad K_{ij} = [u_D z^a] \Psi(q_i) \Psi(q_j)^{-1} (u_E z^b) \quad \text{when } E \mid D, E \neq D, \quad K_{ij} = 0 \quad \text{otherwise.}$$

Lemma 7.9 is an explicit formula for K_{ij} : retain the chains whose deformation labels multiply u_E to u_D and whose exponent increments carry b to a .

Theorem 7.10 (Triangular seed inversion). *On every finite divisor-closed coefficient set, let A_i be the coefficient of slot i at its anchor cut and let κ_i be its canonical seed. In vector form,*

$$(8) \quad A = (1 + K)\kappa, \quad \kappa = (1 + K)^{-1}A = \sum_{n \geq 0} (-K)^n A.$$

In particular,

$$(9) \quad \kappa_i = A_i + \sum_{n \geq 1} (-1)^n \sum_{i_0 < \dots < i_n = i} K_{i_n i_{n-1}} \cdots K_{i_1 i_0} A_{i_0},$$

where $<$ includes strict divisibility of the deformation monomials and the intermediate boundary exponents are summed. Thus every seed is a finite alternating sum of anchored open FJRW coefficients weighted by ordered broken-line chains.

Proof. Evaluate Theorem 1.5 at the anchor of i . The seed in slot i contributes κ_i , since a positive-degree wall acting on it overshoots its deformation monomial. A seed in $j < i$ contributes $K_{ij}\kappa_j$, and no other seed contributes. Hence $A_i = \kappa_i + \sum_{j < i} K_{ij}\kappa_j$. The kernel strictly increases deformation monomials and is nilpotent in the finite quotient, so its geometric series inverse is finite and gives (8)–(9). $\square$

The inversion has a zero-dimensional enumerative lift. Let $\mathcal{Z}_i^{\text{anc}}$ be the signed, automorphism-weighted zero-cycle of the balanced descendent Witten multisection at q_i , so $\deg \mathcal{Z}_i^{\text{anc}} = A_i$. For a term $c_\lambda u_\lambda X_{k_\lambda}$ in a ray logarithm, let $\mathcal{Z}_\lambda^{\text{crit}}$ be the signed weighted union of the zero loci of the corresponding critical-graph homotopies in the staged realization. Exact-support ray purity identifies its degree with c_λ .

For a decorated bend chain γ beginning at $i_0 = (E, b)$, define the rational oriented cascade cycle

$$(10) \quad \mathcal{C}_\gamma = \frac{\prod_h \epsilon_{j(h)} \det(a^{h-1}, k_{\lambda_h} + (1, 1))}{\prod_j n_j!} \left(\mathcal{Z}_{i_0}^{\text{anc}} \times \prod_h \mathcal{Z}_{\lambda_h}^{\text{crit}} \right).$$

Equations (5.36)–(5.37) of [GKT22] identify one such bend with the corresponding critical boundary zero locus: its zero count is the critical-homotopy count times the balanced count, multiplied by precisely the determinant/gluing factor in (10). Associativity of the assembling operation [GKT22, §2.3.2] allows these factors to be iterated. Expanding the kernels along a strict divisibility chain $i_0 < \dots < i_n = i$ therefore gives a finite cascade cycle $\mathcal{C}(i_0 < \dots < i_n = i)$ of degree $K_{i_n i_{n-1}} \dots K_{i_1 i_0} A_{i_0}$.

Theorem 7.11 (Zero-bordism class of the initial coefficients). *The rational zero-cycle*

$$(11) \quad \mathcal{Z}_i^{\text{seed}} = \mathcal{Z}_i^{\text{anc}} + \sum_{n \geq 1} (-1)^n \sum_{i_0 < \dots < i_n = i} \mathcal{C}(i_0 < \dots < i_n = i)$$

is finite and satisfies

$$\deg \mathcal{Z}_i^{\text{seed}} = \kappa_i.$$

Its class in rational oriented zero-bordism is independent of the endpoint representatives, canonical homotopies, and transverse perturbations used in the staged realization.

Proof. Taking degrees in (11) gives the right side of (9), hence κ_i . Anchored degrees are coefficients of the canonical cut potentials, and the critical degrees are coefficients of the canonical ray factors; freeness and ray purity make both independent of the chosen representatives. Finally, rational oriented zero-bordism is classified by weighted degree, $\Omega_0^{\text{or}} \otimes \mathbb{Q} \cong \mathbb{Q}$, which proves independence of the class. $\square$

For later use, $\mathcal{PM}_\Gamma$ denotes GKT's partial compactification of the moduli space associated with a graded graph Γ , and $\partial^+ \mathcal{PM}_\Gamma$ denotes its positive boundary. A finite collection $\mathcal{G}$ of graded graphs is *construction-closed* if it contains every base graph that occurs in the iterated boundary decompositions used to assemble the canonical data of its members. A *P-family* is a multisection of the pullback bundle over $P \times \mathcal{PM}_\Gamma$ which satisfies the canonical boundary identities of [GKT22, Def. 3.22] on every parameter slice.

Lemma 7.12 (Polytope-relative canonical extension). *Let P be a compact smooth polytope with corners and let $\mathcal{G}$ be a finite construction-closed collection of graded graphs. Suppose compatible, automorphism-invariant canonical P -families are prescribed on*

$$(\partial P \times \mathcal{PM}_\Gamma) \cup (P \times \partial \mathcal{PM}_\Gamma)$$

and near $P \times \partial^+ \mathcal{PM}_\Gamma$, for every $\Gamma \in \mathcal{G}$. Assume that the prescribed families are already transverse on every parameter face and graph stratum, that the graph-boundary values are assembled from the corresponding families on the base components, and that they are strongly positive near the positive boundary. They extend, relative to a smaller collar of the prescribed set, over $P \times \mathcal{PM}_\Gamma$, and the extensions may be chosen transverse on every parameter face and graph stratum. Two extensions fixed on the prescribed set are joined by such an extension over $P \times [0, 1]$.

Proof. Repeat the induction on $\dim \mathcal{PM}_\Gamma$ in [GKT22, Lem. 3.40]. Once the lower graph families are fixed, assembling prescribes the section on $P \times \partial \mathcal{PM}_\Gamma$; associativity of assembling [GKT22, §2.3.2] makes the prescriptions agree at graph corners. Orbifold collars and the first three steps of the BCT extension argument used in that proof extend the data while preserving strong positivity. On the compact complement choose finitely many local multisections spanning the fibers, together with collars of every parameter face and graph stratum. Proceed by increasing stratum dimension. A generic small linear combination supported away from the fixed collars is transverse on the current stratum without changing the lower strata; there are only finitely many strata. Taking the finite automorphism orbit at each step preserves the prescribed invariance. The same relative construction over $P \times [0, 1]$, with constant collars at its two ends, proves the final assertion. $\square$

Theorem 7.11 supplies a zero-dimensional rational oriented zero-bordism class for every seed coefficient. The following examples show that these initial coefficients contain information not generated by the ray factors themselves.

Observation 7.13. The open theory is strictly richer than its tropicalization. Broken lines generate exactly the part of the potential reachable by transport from the seeds; the seed system is input that the ray factors do not determine, and it is not zero. This has no analogue in the Fano case of [Gro10, CPS24], where every correction is disk-generated. The witness is Example 7.14: an initial coefficient of value $-\frac{25}{16}$ at the middle balanced point that no ray action can produce.

Example 7.14 (Initial coefficients not generated by wall-crossing). Two things this example shows: bending cannot generate everything, and seeds are not confined to the coefficients that carry no wall. For the first, the invariant $\langle \tau_0^{(3,3)^3} \sigma_1^4 \sigma_2^4 \sigma_{12} \rangle = \frac{1}{5}$ of (3) has no bending source (t -monomials multiply along a broken line, and the unique candidate wall fixes the $t_{3,3}$ seed exactly), so such coefficients are initial data. For the second, take the $N = 2$ one-parameter descendent family $t_{3,3,d=2}$ of $x^5 + y^5$, the exact solution of Theorem 7.7 has equal wall coefficients $c_1 = c_2 = \frac{5}{144}$, extreme value $\langle \tau_2^{(3,3)} \sigma_1^3 \sigma_2^{13} \sigma_{12} \rangle = \frac{25}{36}$, and initial coefficient at the middle balanced point $\langle \tau_2^{(3,3)} \sigma_1^8 \sigma_2^8 \sigma_{12} \rangle^{\text{mid}} = -\frac{25}{16}$, the $\mathcal{A}$ -condition holding in all three chambers.

Example 7.15 (An anchored value need not be an initial coefficient). Theorem 7.5 excludes foreign contributions only from beyond the target's own direction; contributions from the $(1, 1)$ -side may remain at the anchor cut. Composite diagonals with descendents exhibit this behavior. In the smallest off-ray example, the anchored value is 0 while the initial coefficient is $-\frac{10}{3}$, their difference being a foreign contribution at the anchor cut. In detail, $J = \{(0, 0, 2), (1, 1, 0)\}$ and the off-ray $J = \{(0, 0, 2), (2, 1, 0)\}$ are solved completely (§8): the unit-twist tower $t_{0,0,2}$ carries

nonzero wall coefficients $\frac{5}{6}$ on both of its near-axis rays, the composite diagonals have forced own wall coefficients 1, and the $\mathcal{A}$ -conditions hold in every chamber of the finite family. In the off-ray case the foreign contribution contributes $\frac{10}{3}$ to the *extreme* coefficient $(7, 1)$ in its anchor cut: the anchored value $(0, \text{the normal form})$ is not the seed, and the correct object is canonical.

7.3. Realization by staged homotopies.

Proposition 7.16 (Descendent rays accumulate; primary rays are locally finite). *Fix $r, s \geq 2$ and an interior twist (a, b) . Let d range over $\mathbb{Z}_{\geq 1}$, so that $t_{a,b,d}$ runs over deformation parameters of total degree one. The rays of the associated critical graphs are exactly*

$$R(a, b) = \{(a + jr + 1, b + \kappa s + 1) : j, \kappa \in \mathbb{Z}_{\geq 0}\}, \quad d = j + \kappa + 1,$$

their wall functions in $\mathfrak{D}^{r,s}$ are all nonzero, and their slopes are dense in $(0, \infty)$. Hence the walls of $\mathfrak{D}^{r,s}$ accumulate in every subsector of the direction sector already at total degree one in the deformation parameters, for every (r, s) , the simple singularities included; those carry no primary wall at all (Theorem 1.1). By contrast, write $\mathfrak{D}_0^{r,s}$ for the primary boundary-ray factorization, the factors with $\mathbf{d} = \mathbf{0}$, and $\mathfrak{m}$ for the ideal generated by the deformation parameters. Then $\mathfrak{D}_0^{r,s}$ has finitely many walls modulo $\mathfrak{m}^n$ for every n , and its chambers are open angular sectors.

Proof. For a singleton $J = \{(a, b)\}$ one has $\ell_1 = \ell_2 = 0$ and $N = d$, so by Definition 2.1 the critical graphs $\Lambda_{J,p}$, $1 \leq p \leq d$, have boundary degrees $(a + (p-1)r + 1, b + (d-p)s + 1)$; putting $j = p-1$, $\kappa = d-p$ gives $j + \kappa = d-1$ and the stated set, every element of which is attained by taking $d = j + \kappa + 1$. For density, fix $t > 0$ and let j be large enough that $t(a + jr + 1) > 1 + b$; with $\kappa = \lceil (t(a + jr + 1) - 1 - b)/s \rceil$ one has $\kappa s \in [t(a + jr + 1) - 1 - b, t(a + jr + 1) - 1 - b + s)$, whence

$$\left| \frac{b + \kappa s + 1}{a + jr + 1} - t \right| < \frac{s}{a + jr + 1} \xrightarrow{j \rightarrow \infty} 0.$$

For the wall functions, note that on a singleton diagonal the chamber-index condition is $\mathcal{A}(J, \mathbf{d}, \nu) = (-1)^d \neq 0$, whose $h = 1$ part is $\sum_p \omega_p \nu_p$ with all $\omega_p > 0$, so $\nu_N^{\max} \neq 0$; solving $W^{\nu^{\max}} - W^{\nu^{\min}} = \sum_p c_p w_p$ with $w_p = a_p e_p - b_p e_{p-1}$, $a_p, b_p > 0$, gives $c_N = \nu_N^{\max}/a_N \neq 0$ and $c_p = c_{p+1} b_{p+1}/a_p \neq 0$ for every p , and no cancellation between distinct rays is possible at total degree one, where the monomials are distinct. For the primary statement, modulo $\mathfrak{m}^n$ only monomials $u_{J,0}$ with $|J| < n$ occur, and for each such $|J|$ the twists range over the finite Neveu–Schwarz box, so only finitely many diagonals, hence finitely many walls, contribute; finitely many rays through the origin cut the sector into open angular sectors. $\square$

In the statement and proof of the next theorem, a jump, stage, or degree *survives the quotient* if its class modulo that quotient is nonzero.

Theorem 7.17 (Geometric realization in a finite coefficient quotient; = Theorem 1.6). *Let S be a finite divisor-closed set of deformation monomials containing 1, and choose a finite labeled set I with at least $n_{a,b}(S)$ labels of each twist occurring in S , with $n_{a,b}(S)$ as in Theorem 1.6. The quotient factorization has finitely many walls, with rays $\rho_1, \dots, \rho_n$ listed in increasing slope. There is a canonical homotopy $(\mathfrak{s}_t)_{t \in [0,1]}$, obtained by concatenating families of simple canonical homotopies in the sense of [GKT22, §3.4], from a family of canonical multisections realizing $\nu_I^{\min}$ to one realizing $\nu_I^{\max}$, the restrictions of the universal extreme indices, such that after reduction to the quotient supported on S every nonzero jump is supported on a single boundary ray, the surviving rays occur in increasing slope as t increases, and the total reduced jump on ρ_i is the wall automorphism θ_{ρ_i} of Theorem 1.3 modulo that quotient.*

Proof. Step 1: the labeled model, and symmetrization. Let $\mathfrak{q}_S$ be the monomial ideal spanned by monomials outside S ; it is an ideal because the complement of a divisor-closed set is upward closed. Write a bar for reduction modulo $\mathfrak{q}_S$. Let $A_I = \mathbb{Q}[u_{i,j} : i \in I, j \geq 0] / (u_{i,j} u_{i,j'})$ be the labeled coefficient ring of [GKT22, Def. 4.6], square-zero in each label; write $u_{A,\mathbf{d}} = \prod_{i \in A} u_{i,d_i}$ and $u_i = u_{i,0}$. The monomials $u_{J,\mathbf{d}}$ (indexed by a twist multiset, in the deformation parameters) and $u_{A,\mathbf{d}}$ (indexed by a label subset, in A_I) live in different rings. Symmetrization sends, for every descendent level,

$$t_{a,b,d} \mapsto \sum_{\substack{i \in I \\ \text{tw}(i)=(a,b)}} u_{i,d}.$$

Let $\mathfrak{q}_{I,S}$ be the labeled monomial ideal whose monomials have symmetric type outside S . Lemma 4.14 of [GKT22] says that symmetrization induces an injection

$$\mathbb{Q}[t_{a,b,d}]/\mathfrak{q}_S \hookrightarrow A_I/\mathfrak{q}_{I,S}.$$

Indeed, a monomial in S maps to the nonempty sum over injective assignments of its occurrences to labels of the required twists; the saturation bound $n_{a,b}(S)$ makes the sum nonempty, and distinct symmetric monomials have disjoint labeled supports. The induced map on the enlarged Lie algebra preserves shifted boundary rays, including the marginal $(1,1)$ -ray. The action after reduction is still free: the minimal-support argument of Lemma 2.5 detects every nonzero class supported in S on W_0 . Thus the $\mathfrak{q}_{I,S}$ -reductions of $\nu_I^{\min}, \nu_I^{\max}$ and of the labeled boundary-ray factorization identify with their universal symmetric counterparts. Below we suppress the labeled-lift symbol in the action of a symmetric factor.

Step 2: the ray factors are symmetric. Twist-preserving permutations fix the two symmetric extreme indices, hence by freeness fix $g^{\min \rightarrow \max}$; they also preserve every boundary-ray subspace. Uniqueness of the factorization in Theorem 1.2(ii), whose filtered proof applies verbatim over the labeled coefficient ring, therefore makes each θ_{ρ_i} symmetric. Write the boundary-ray factorization, in chronological convention, as

$$g^{\min \rightarrow \max} = \theta_{\rho_n} \circ \cdots \circ \theta_{\rho_1} \pmod{\mathfrak{q}_{I,S}}, \quad \text{slope}(\rho_1) < \cdots < \text{slope}(\rho_n).$$

Step 3: one stage per ray. For each quotient factor choose the symmetric full-group representative $\widehat{\theta}_{\rho_i} = \exp(\widehat{\eta}_{\rho_i})$ whose logarithm is the finite sum of its terms supported in S and has no terms outside S . Starting with a transverse symmetric canonical family having index $\nu_I^{\min}$, construct one stage for each factor. If the current index is ν , then $\widehat{\theta}_{\rho_i} \nu$ is again a symmetric element of $\Omega_{r,s}(I)$. The realizability statement $\Omega_{r,s}(I) = \Upsilon_{r,s}(I)$ ([GKT22, Cor. 5.14], together with [GKT22, Prop. 3.46]), read in the enlarged sense of Lemma 2.5 on the marginal ray, supplies a symmetric canonical endpoint family with this index. The constructive realization in [GKT22, Thm. 5.13, Lem. 5.15] may be chosen transverse, and twist symmetrization is performed branchwise, preserving transversality. Join the two transverse endpoint families by a family of simple canonical homotopies as in [GKT22, Lem. 3.40].

Step 4: ray purity of each stage. We claim that every jump internal to this stage is supported on ρ_i . For the actual chronological order of its subset times, [GKT22, Thm. 5.9] writes the total labeled jump as

$$\prod_{\substack{\text{time} \\ \emptyset \neq A \subseteq I}} \exp(v_A),$$

where v_A has exact label support A (it is the sum over descendent profiles of terms $u_{A,\mathbf{d}} X$). In the labeled coefficient ring, $u_{A,\mathbf{d}} u_{B,\mathbf{e}} = 0$ when $A \cap B \neq \emptyset$. For any fixed order this exact-support factorization is unique. Indeed, a nonzero bracket of disjoint supports has support

$A \cup B$; hence

$$\log \left(\prod_A \exp(v_A) \right) \Big|_A = v_A + P_A(v_B : B \subsetneq A),$$

and induction on $|A|$ recovers every v_A .

The total jump and the labeled lift of $\widehat{\theta}_{\rho_i}$ carry the same initial endpoint potential to the same final one. Freeness of the enlarged labeled action (Lemma 2.5) makes them equal. Now induct again on $|A|$. The exact- A component of the logarithm of the target is supported on ρ_i ; by induction all v_B for $B \subsetneq A$ are supported there as well. Every Lie word in P_A therefore vanishes because the boundary-ray subalgebra is abelian, as proved in Theorem 1.2(ii). Thus v_A equals the target's exact- A component and is supported on ρ_i . This proves the claim, including for the enlarged torus factor.

Step 5: the correction stage. After the stages that survive the quotient, the resulting full index ν^* is congruent to $\nu_I^{\max}$ in the stated quotient. Let g_* be the unique full transition from ν^* to $\nu_I^{\max}$. Quotient freeness gives $\bar{g}_* = \text{id}$. Join the current family to a transverse symmetric family realizing $\nu_I^{\max}$ by one further Lemma-3.40 stage, chosen as above. Its total jump equals g_* by full freeness. Applying the exact-support induction above *after reduction* to its time-ordered factorization shows successively that every one of its internal jump factors has trivial class: this correction stage is entirely zero modulo $\mathfrak{q}_{I,S}$.

Step 6: concatenation. Give all stages constant collars at their endpoints, reparametrize them to successive subintervals of $[0, 1]$, and concatenate. Canonicity, strong positivity and simplicity ([GKT22, Defs. 3.18, 3.22, 3.36–3.37]) survive this operation: use identical collars and time reparametrizations simultaneously on every graph and component, which preserves the boundary-factorization identity [GKT22, (3.12)] and the component-independence defining simplicity; transversality is unchanged away from the collars and follows on them from spatial transversality of the endpoint sections, while strong positivity is slicewise. The resulting homotopy has the exact endpoints and the asserted reduced order and total jumps. It is not one global Lemma-3.40 normal-form family: the same marking subset may jump in different ray stages. That distinction is precisely what evades Proposition 7.21. $\square$

Remark 7.18 (Finite versus completed realization). Theorem 7.17 is a theorem for each fixed finite quotient. The ray factors are algebraically compatible when the coefficient window grows, but the proof does not choose their endpoint families and homotopies compatibly. Such a choice would require relative lifting theorems for the spaces of canonical families and simple canonical homotopies. We therefore do not assert a single completed geometric homotopy for the dense descendent factorization. Moreover, the final correction stage is zero in the stated quotient but may have jumps on several rays outside it.

A single unstaged homotopy already suffices below the first degree at which one diagonal carries two walls.

Proposition 7.19 (Exact first two-wall order; unstaged realization). *For $m \geq 1$, the maximum of the numerical diagonal count $N(J, \mathbf{0})$ over primary multisets with $|J| = m$ is*

$$N_{\max}(r, s; m) = m + 1 - \left\lceil \frac{2m}{r} \right\rceil - \left\lceil \frac{2m}{s} \right\rceil,$$

attained by $J = \{(r-2, s-2)^m\}$. Consequently the least primary degree admitting a diagonal with two walls is

$$N_2(r, s) = \min \left\{ m \geq 3 : \left\lceil \frac{2m}{r} \right\rceil + \left\lceil \frac{2m}{s} \right\rceil \leq m-1 \right\},$$

where the minimum of the empty set is ∞ .

- (i) $N_2 = \infty$ if and only if $\hat{c}_W \leq 1$, that is, if and only if W is simple or parabolic; when $\hat{c}_W > 1$, $N_2 \geq \lceil 1/(\hat{c}_W - 1) \rceil$.
- (ii) When $N_2 < \infty$, one unstaged Lemma-3.40 normal-form homotopy realizes the primary boundary-ray factorization modulo $\mathbf{m}^{N_2}$. When $N_2 = \infty$, such a homotopy exists at every finite order, realizing the completed primary boundary-ray factorization as the inverse system of these finite-order statements; no compatibility of the homotopies themselves is asserted.

Proof. Write $\sum_j a_j = r(J) + r\ell_1$ and $\sum_j b_j = s(J) + s\ell_2$. Since $a_j \leq r-2$,

$$\ell_1 \leq \left\lfloor \frac{m(r-2)}{r} \right\rfloor = m - \left\lceil \frac{2m}{r} \right\rceil,$$

and similarly for ℓ_2 ; both bounds are attained simultaneously by the maximal twist. Substitution in $N = \ell_1 + \ell_2 - m + 1$ proves the formula for $N_{\max}$ and hence the displayed formula for N_2 . In particular $m = 1, 2$ never gives $N \geq 2$.

The coarser bound from the proof of Theorem 1.1 is $\ell_1 + \ell_2 \leq m\hat{c}_W$, hence

$$(12) \quad N(J, \mathbf{0}) \leq 1 + m(\hat{c}_W - 1),$$

which is at most 1 for every m when $\hat{c}_W \leq 1$, and, when $\hat{c}_W > 1$, forces $m \geq 1/(\hat{c}_W - 1)$ whenever $N \geq 2$; this is (i) apart from the converse. Conversely, if $\hat{c}_W > 1$, the exact maximal-twist count satisfies $N_{\max} > m(\hat{c}_W - 1) - 1$, which is at least 2 for all sufficiently large m , so $N_2 < \infty$.

For (ii), put $K = N_2$ when $N_2 < \infty$ and otherwise fix an arbitrary finite K . Work modulo $\mathfrak{q}_K = (\mathbf{m}^K, \text{descendent parameters})$ and choose at least $K-1$ labels of every primary twist, so the symmetrization map used in Theorem 7.17 is injective in the degrees retained by the quotient. Then every retained diagonal has $N \leq 1$, and a marking subset $A \subseteq I$ determines the single primary diagonal $(A, \mathbf{0})$. Choose transverse symmetric families realizing $\nu_I^{\min}$ and $\nu_I^{\max}$ and take the one Lemma-3.40 normal-form family between them. Its jumps occur at the subset times t_A . By the element displayed in the proof of [GKT22, Thm. 5.9], the reduction of the jump at t_A is a sum over the critical graphs with marking set A : it is zero if $|A| \geq K$ or no retained wall exists, and otherwise is supported on the unique wall of that diagonal.

Order the times of all subsets whose retained primary diagonal carries its unique wall by that wall's slope; place the no-wall and discarded subset times arbitrarily. A resulting zero jump is harmless. This prescription is legitimate: [GKT22, Lem. 3.40] constrains the times only to be distinct in $(0, 1)$, and its construction takes them as given: the zeros are moved onto a prescribed $t_{I(\Gamma)}$ by a diffeomorphism preserving the moduli factor, and the finitely many non-transverse times are kept off the t_A by re-extending transversally at each prescribed time. Visible jumps sharing a ray commute by the abelianness proved in Theorem 1.2(ii). Full freeness identifies the total product with $g^{\min \rightarrow \max}$; after reduction, its grouping by ray is therefore the unique boundary-slope factorization $\prod_{\rho}^{\rightarrow} \theta_{\rho}$. Symmetry and injectivity descend this labeled statement to the universal primary quotient. Thus the total reduced jump on ρ_i is θ_{ρ_i} . $\square$

Remark 7.20. The ordering step above is the only place the freedom in the times of [GKT22, Lem. 3.40] is used, and the freedom is real rather than assumed: the lemma imposes on the family $(t_A)_{A \subseteq I}$ only that its members be distinct points of $(0, 1)$, its proof never constructs a

t_A but consumes each as a given target, and the induction there runs on $\dim \mathcal{PM}$, which is unrelated to the marking set indexing the times. What the ordering does *not* do is separate the walls of a single diagonal: those all carry the one marking set J , so they share the one time t_J whatever order is chosen, which is Proposition 7.21.

At the first two-wall degree, one global Lemma-3.40 normal-form family provably does not suffice. This explains why the stagewise construction in Theorem 7.17 is necessary.

Proposition 7.21 (One normal-form family cannot separate a diagonal's walls). *Let $(J, \mathbf{0})$ be a primary diagonal with $N \geq 2$ and let H^* be a family of simple canonical homotopies with times $(t_A)_{A \subseteq I}$ as in [GKT22, Lem. 3.40]. Then every possible contribution from the diagonal's N walls is concentrated at the single time t_J , with wall p actually contributing if and only if $\Delta(\Lambda_{J,p}) \neq 0$. The jump there is supported on one ray if and only if at most one of these zero counts is nonzero. In particular no single such normal-form family gives the ray-by-ray realization of Theorem 7.17 once two of the diagonal's walls are actually crossed.*

Proof. The N critical graphs $\Lambda_{J,1}, \dots, \Lambda_{J,N}$ all carry the internal marking set J , so by [GKT22, Lem. 3.40(3)(a)] their zeros occur at the one time t_J . By the jump formula in the proof of [GKT22, Thm. 5.9], the logarithm of the jump there is

$$\sum_{p=1}^N (-1)^{|J|+k_2(\Lambda_{J,p})-1} \Delta(\Lambda_{J,p}) u_{J,\mathbf{0}} X_{k_1(\Lambda_{J,p})-1, k_2(\Lambda_{J,p})-1}.$$

Its boundary directions are pairwise distinct: along p the first boundary degree increases by r and the second decreases by s , both by at least 2 from values at least 1, so the slope strictly decreases; and for $N \geq 2$ none of them is the torus direction. The displayed generators lie in distinct bidegree summands, so no cancellation is possible. Its logarithm is therefore supported on one ray exactly when at most one coefficient is nonzero. $\square$

The stagewise theorem lies outside this family: it applies Lemma 3.40 separately to each boundary-ray factor and may therefore reuse the same marking subset at several stage times. Proposition 7.19 succeeds without staging because when $N \leq 1$ the at-most-one condition on $\Delta(\Lambda_{J,p})$ holds for free.

Example 7.22 (The first two-ray diagonal of $x^5 + y^5$). For $x^5 + y^5$ the order $N_2 = 5$ of Proposition 7.19 is attained by $J = \{(3, 3)^5\}$, $\mathbf{d} = \mathbf{0}$: here $m(J, \mathbf{0}) = 50$, the Euler weight is 25, and $N = 2$, with balanced profiles $(0, 10), (5, 5), (10, 0)$ and critical graphs of boundary degrees $(1, 6)$ and $(6, 1)$: two walls on two *distinct* rays, the first such diagonal for this Fermat pair. Their jump vectors are $w_1 = 30e_1 - 5e_0$ and $w_2 = 5e_2 - 30e_1$. Since $d(J, \mathbf{0}) = -3 < 0$ the chamber-index condition is $\mathcal{A}(J, \mathbf{0}, \nu) = 0$, which reads

$$\frac{6}{25} \nu_{(0,10)} + \frac{1}{25} \nu_{(5,5)} + \frac{6}{25} \nu_{(10,0)} = \frac{204}{15625} :$$

three coefficients against two torsor directions and one condition, as Theorem 1.4 requires. In the minimal normal form of Theorem 1.3 this gives

$$\langle (\tau_0^{(3,3)})^5 \sigma_2^{10} \sigma_{12} \rangle^{\min} = \frac{34}{625},$$

with $\langle (\tau_0^{(3,3)})^5 \sigma_1^{10} \sigma_{12} \rangle^{\max}$ the same by symmetry. (Its agreement with a closed value of Example 6.1 is a numerical accident.) This is the lowest primary degree for $x^5 + y^5$ at which a single unstaged jump would have to carry two rays, hence the smallest test distinguishing Theorem 7.17 from one global Lemma-3.40 normal-form family; every sector computed in §5 and in §7 below has $N \leq 1$, which is why the difficulty does not surface there.

The hypothesis of Proposition 7.21 is met here. Take I to carry exactly five markings of twist $(3, 3)$, so that this monomial is realized by a single marking set and the walls above it share one time. Writing $\nu^{\max} - \nu^{\min} = c_1 w_1 + c_2 w_2$ and reading the extreme rows gives $c_1 = c_2 = \frac{34}{3125}$, the middle row being consistent; more generally $c_1 = \nu_0/b_1$ and $c_{p+1} = a_p c_p/b_{p+1}$, so every c_p is nonzero as soon as ν_0 is, exactly as in the proof of Proposition 7.16. No lower-order term can absorb either coefficient: the walls of this tower sit at $|J| = 2, 4, 5$ and no multiset drawn from the lower orders $|J| = 2, 4$ sums to 5, while a wall acting on a lower coefficient of the potential contributes nothing here, its direction being parallel to that coefficient's and hence annihilating it (Proposition 7.2). So both zero counts are nonzero and both walls are crossed at the one time.

The same computation shows the difficulty is not peculiar to the hyperbolic regime. A single insertion of maximal twist with two descendents has $N = 2$ for every (r, s) , with both coefficients again nonzero, so the obstruction appears in every model once descendents are admitted, the simple ones included, which carry no primary wall at all. What is special about $x^5 + y^5$ is that it is the first symmetric hyperbolic Fermat pair and this is its first primary occurrence.

Corollary 7.23. *For $x^5 + y^5$ modulo $\mathfrak{m}^6$, no family of simple canonical homotopies in one global normal form supplied by [GKT22, Lem. 3.40] between families realizing the extreme indices $\nu^{\min}$ and $\nu^{\max}$ has all of its jumps supported on single rays.*

Proof. By Example 7.22 both walls of the diagonal $J = \{(3, 3)^5\}$ are crossed, and by Proposition 7.21 they share the single time t_J , so the jump there meets two distinct rays. $\square$

Remark 7.24. For simple W the *primary* statement carries no walls at all: $N \leq 0$ throughout by Theorem 1.1, so $\mathfrak{D}_0^{r,s}$ is empty and the direction sector is a single chamber. By Proposition 7.16, though, the descendent walls of the same model are dense in that same sector. The parabolic pairs are the first case with primary content, and there the unique ray is $(1, 1)$, carrying the marginal torus direction; the jump it effects is the one exhibited in [GKT26, Ex. 4.9]. In the hyperbolic regime diagonals with $N \geq 2$ occur and one unstaged jump can then carry several rays at once. The exact first degree is given by Proposition 7.19. Up to the symmetry $(r, s) \leftrightarrow (s, r)$, its complete hyperbolic classification is

r	s	$N_2(r, s)$
3	7	21
3	8	12
3	$9 \leq s \leq 11$	9
3	$s \geq 12$	6
4	5	10
4	$6 \leq s \leq 7$	6
4	$s \geq 8$	4
5	$5 \leq s \leq 7$	5
5	$s \geq 8$	4
$r \geq 6$	$s \geq r$	3

Thus $N_2 \leq 21$ for hyperbolic integral Fermat exponents, with the maximum attained only at $(3, 7)$ up to symmetry. The coarser lower bound in Proposition 7.19 is useful for a fixed pair but does not produce an unbounded sequence approaching the threshold: the exponents are discrete.

Remark 7.25 (Why staging is necessary). On every finite divisor-closed coefficient set, Theorem 7.17 realizes the successive slope cuts by intervals in $[0, 1]$, the two sides of the direction sector by the two endpoints, and the extreme chamber indices by $\nu^{\min}$ and $\nu^{\max}$. In a finite primary quotient these cuts bound actual open chambers; in the descendent completion only the algebraic ray factors are asserted to be coefficientwise compatible. The marginal torus factor of Theorem 1.2(ii) is carried by the $(1, 1)$ -ray, whose fixed monomials are exactly its own.

Proposition 7.21 does not contradict the theorem: it concerns one global Lemma-3.40 normal-form family, in which a marking subset has only one jump time. The proof of the theorem instead connects the endpoints of each boundary-ray factor in a separate stage. Exact labeled-support uniqueness forces every internal jump of that stage onto the prescribed ray, and concatenation may reuse a marking subset in later stages. The ray-purity step does not use the single-critical-graph modification of [GKT22, Lem. 5.17] to prescribe a stage jump, nor does it identify that lemma's relative Euler number with the zero count in [GKT22, Thm. 5.9]. (The established endpoint-realizability theorem [GKT22, Cor. 5.14] remains part of the input.)

7.4. The punctured boundary-charge sector. The open FJRW invariants of $x^r + y^s$ are coefficientwise weighted broken-line transports on the direction sector, with the closed extended invariants of §6 as vertex data. Theorem 7.11 gives the seed constants an enumerative zero-cycle class. We now determine exactly how much position-space geometry the intrinsic boundary-charge sector of Definition 7.1 supports and prove the obstruction to the stronger interpretation.

We now turn to the position-space question. The lattice, cone, axes, ray placement, determinant pairing and primitive-form shift are intrinsic boundary-charge data. The lattice even supplies natural affine normalizations, for example total boundary charge. The issue is not the existence of a coordinate called radius, but whether continuation detects it and whether the cone supports the required singular affine monodromy.

Put $C_{\partial}^{\times} = C_{\partial} \setminus \{0\}$. For a finite divisor-closed coefficient set S , write $\mathfrak{D}_S$ for the finite union of wall rays in the quotient. A *tame path* has off-wall endpoints and is piecewise smooth and transverse to $\mathfrak{D}_S$, apart from finitely many generic tangencies; tame homotopies are taken relative endpoints and avoid the deleted vertex.

Theorem 7.26 (Projective direction sector and obstruction to a singular integral-affine base; = Theorem 1.7). *Let S be finite and divisor-closed.*

- (1) *The wall-stratified sector is isomorphic to $(0, \infty) \times [0, 1]$, with every wall equal to a radial half-line times a marked point. It admits a stratum-preserving strong deformation retraction to a constant-radius interval.*
- (2) *The crossing word of a tame path is determined by its angular projection. A tame homotopy inserts or deletes only adjacent inverse crossings, and every crossing word reduces uniquely to the monotone path in the chamber-adjacency tree.*
- (3) *Choose the intermediate families and ray stages supplied by Theorem 7.17. Assigning $\theta_{\rho, S}$ to a positive crossing of ρ and $\theta_{\rho, S}^{-1}$ to a negative crossing gives a reduced automorphism depending only on the angular endpoint chambers. A backtracking cancellation is realized by a relative two-parameter canonical family, unique up to a relative three-parameter family. If*

$$\pi : C_{\partial}^{\times} \longrightarrow \mathbb{P}(C_{\partial}) \cong [0, 1]$$

this reduced continuation observable factors through π : radial paths act by the identity, and positive dilation changes neither crossing words nor their reduced automorphisms.

- (4) *The punctured sector contains no loop around the deleted vertex and no joint that compares the two orders of distinct rays. The total product $g^{\min \rightarrow \max}$ is boundary-to-boundary transport, not holonomy about the vertex.*

Consequently the continuation structure constructed here does not by itself supply a radially detected modulus or a singular affine origin with monodromy. In particular the marginal torus factor is not thereby identified with integral-affine holonomy. A Gross–Siebert/SYZ position space requires additional input.

Proof. Write $q = x_1 e_1 + x_2 e_2$ and set

$$R(q) = x_1 + x_2, \quad a(q) = \frac{x_2}{x_1 + x_2}.$$

Then

$$(13) \quad (R, a) : C_\partial^\times \xrightarrow{\sim} (0, \infty) \times [0, 1], \quad (r, t) \mapsto r((1-t)e_1 + te_2).$$

If a_i is the slope coordinate of a wall ρ_i , then $\rho_i \setminus \{0\} = (0, \infty) \times \{a_i\}$. Thus the finite wall stratification is the product of a radial half-line with a marked interval. The homotopy

$$(r, t) \mapsto ((1-u)r + u, t), \quad 0 \leq u \leq 1,$$

is a stratum-preserving strong deformation retraction to $R = 1$.

The crossing word of a tame path is therefore the crossing word of its angular projection. Under a generic homotopy the word changes only by inserting or deleting $e_i e_i^{-1}$ or $e_i^{-1} e_i$: a tangency to the i th marked slope creates or cancels that pair. Crossings of distinct slopes cannot exchange order. Equivalently, the chamber-adjacency graph is a finite path, hence a tree, and every word reduces uniquely to the monotone edge path between its endpoint chambers.

Choose from Theorem 7.17 the intermediate canonical families $\mathfrak{s}_0, \dots, \mathfrak{s}_n$ and the ray stages $h_i : \mathfrak{s}_{i-1} \rightarrow \mathfrak{s}_i$. Assign $\mathfrak{s}_i$ to the i th chamber, h_i to e_i , h_i with reversed parameter to e_i^{-1} , and the corresponding product of wall automorphisms to a crossing word. The loop $h_i * h_i^{-1}$ has compatible canonical data on the boundary of a parameter polygon. Lemma 7.12 fills it relative to that boundary. Applying the same lemma over a three-dimensional parameter polytope joins two such fillers relative to their common boundary. Thus adjacent inverse factors cancel both algebraically and through relative canonical families.

The resulting reduced automorphism uses only the angular endpoint chambers: the chamber-adjacency graph is a path, and its unique reduced word is the monotone word between them. A radial path has constant projection and $\pi \delta_\lambda = \pi$ for every positive dilation δ_λ . This proves the first three assertions.

Finally, (13) makes $C_\partial^\times$ contractible. Its two boundary sides are not identified, so the slope-ordered path from the e_1 -side to the e_2 -side is not a loop. Distinct walls are disjoint after deletion of the vertex, and the crossing-word argument shows that the available disks cannot compare their two orders. To create an origin loop one must enlarge the sector to a surface with an interior puncture or glue its two sides. Either operation requires affine transition data, including a linear part in $\mathrm{GL}(N_\partial)$, which the formal wall automorphisms do not supply. This proves the last assertion and the conclusion. $\square$

The theorem resolves the intrinsic version of the position-space question negatively while retaining its positive finite-quotient content: its wall crossing observable is projective. In a genuine tropical Landau–Ginzburg model, the input includes a singular integral-affine manifold, a polyhedral decomposition and a multivalued convex piecewise-linear function, and broken lines use that actual affine geometry [CPS24]. None of those radial or vertex data is recovered by the sector calculation above.

The same boundary applies to theta functions. The present results transport potentials and their seed decomposition, but they do not construct the binary interacting-broken-line or pair-of-pants operation needed to identify theta multiplication constants with the open invariants. Such theta functions (in the sense of [GHK15, GS22]) could still span the Saito–Givental deformation of $x^r + y^s$ and recast the open mirror theorem [GKT22, Thm. 0.3] as intrinsic mirror symmetry. Theorem 7.26 does not rule out that enriched theory; it proves that it requires new input, for example a Novikov/energy valuation, a period normalization, or a logarithmic degeneration with specified affine gluing. The suggested identification of the marginal torus with monodromy of the elliptic family underlying Maher’s modularity [Mah24] is likewise an additional comparison theorem, not a consequence of the present sector.

8. EXACT FINITE CALCULATIONS

The proofs of the structural theorems are independent of software. Exact calculations provide the following reproducibility checks; the repository <https://github.com/bwuebben/open-fjrw-scattering> contains the programs and a file-by-file map from these mathematical statements to their implementations.

Critical graphs and boundary rays. Direct enumeration from [GKT22, Not. 2.40] agrees with the closed formula for $N(J, \mathbf{d})$, the weight trichotomy, the values of $N_{\max}(r, s; m)$ and $N_2(r, s)$, and the two-ray example $J = \{(3, 3)^5\}$. The same calculation gives the ray set $R(a, b)$ of Proposition 7.16. Exact symbolic differentiation verifies bracket closure and the $(1, 1)$ exponent shift.

Periods, closed correlators, and recursion. Expansion of the oscillatory integral gives the seed relation (4) and the Gamma functional of Lemma 4.1. The closed correlators printed in Example 6.1 are obtained independently from Saito flat coordinates, genus-zero tautological intersection theory, and Chiodo’s formula for Witten’s r -spin class. With these values, both sides of [GKT22, Thm. 0.5] agree in all 72 admissible cases with two internal markings and all 105 cases with three.

Ray factors and transport. Exact expansion through deformation degree four reproduces the displayed factors of Theorem 1.3, including $c_{(1,2)} = -1/750$, $c_{(0,2)} = -1/1250$, and the vanishing fourth-order coefficient on the $(1, 1)$ ray. The transport identities reproduce (3), the initial coefficient $-25/16$ in Example 7.14, the foreign contribution $10/3$ in Example 7.15, and the finite chain formula (9).

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